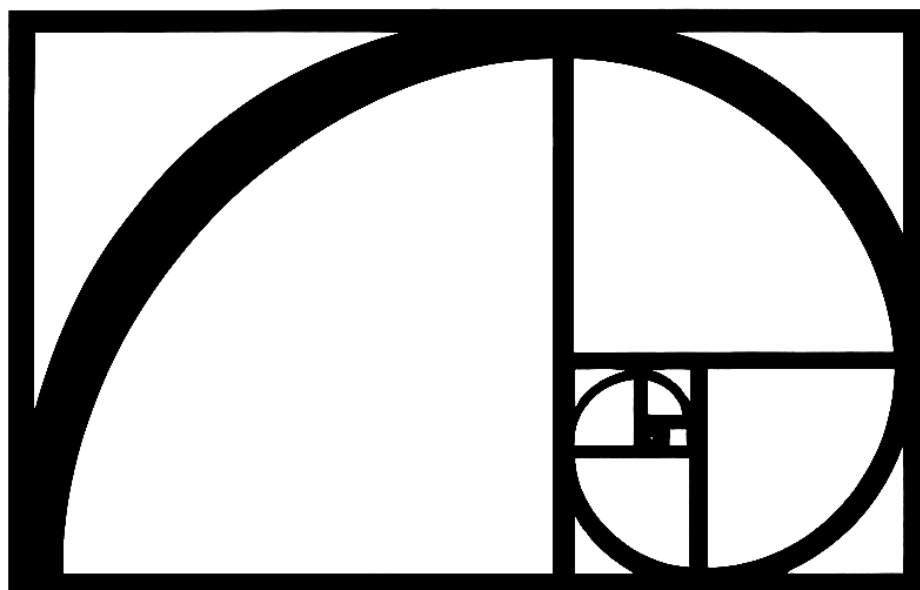




Curiosity 1.0

**24 real exam passages
for curious minds**

Compiled by Temur Normuminov

Introduction

Welcome to *Curiosity 1.0* — a fun reading journey through the surprising history of everyday things that have appeared in real IELTS exams. From pencils and tea to guitars, telegraphs, and even ice cream, each passage is designed to strengthen your reading skills while keeping you curious and entertained.

This book is specially created for those who are just starting IELTS preparation and want to improve reading speed, comprehension, vocabulary, and confidence through interesting real-style texts rather than boring practice.

I hope these short stories inspire you to read more, learn more, and stay curious throughout your IELTS journey.

Sincerely,

Temur Normuminov

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Test 1: The History of Pencil

A

The beginning of the story of pencils started with a lightning. Graphite, the main material for producing pencil, was discovered in 1564 in Borrowdale in England when a lightning struck a local tree during a thunder. Local people found out that the black substance spotted at the root of the unlucky tree was different from burning ash of wood. It was soft, thus left marks everywhere. Chemistry was barely out of its infancy at the time, so people mistook it for lead, equally black but much heavier. It was soon put to use by locals in marking their sheep for signs of ownership and calculation.

B

Britain turns out to be the major country where mines of graphite can be detected and developed. Even so, the first pencil was invented elsewhere. As graphite is soft, it requires some form of encasement. In Italy, graphite sticks were initially wrapped in string or sheepskin for stability, becoming perhaps the very first pencil in the world. Then around 1560, an Italian couple made what are likely the first blueprints for the modern, wood-encased carpentry pencil — their version was a flat, oval, more compact type of pencil. Their concept involved the hollowing out of a stick of juniper wood. Shortly thereafter in 1662, a superior technique was discovered by German people: two wooden halves were carved, a graphite stick inserted, and the halves then glued together – essentially the same method in use to this day. The news of usefulness of these early pencils spread far and wide, attracting the attention of artists all over the known world.

C

Although graphite core in pencils is still referred to as lead, modern pencils do not contain lead as the “lead” of the pencil is actually a mix of finely ground graphite and clay powders. This mixture is important because the amount of clay content added to the graphite depends on intended pencil hardness, and the amount of time spent on grinding the mixture determines the quality of the lead. The more clay you put in, the higher hardness the core has. Many pencils across the world, and almost all in Europe, are graded on the European system. This system of naming used B for black and H for hard; a pencil’s grade was described by a sequence or successive Hs or Bs such as BB and BBB for successively softer leads, and HH and HHH for successively harder ones. The standard writing pencil is graded HB.

D

In England, pencils continued to be made from whole sawn graphite. But with the mass production of pencils, they are getting drastically more popular in many countries with each passing decade. As demands rise, appetite for graphite soars. According to the United States Geological Survey (USGS), world production of natural graphite in 2012 was 1,100,000 tonnes, of which the following major exporters are: China, India, Brazil, North Korea and Canada. When the value of

graphite was realised, the mines were taken over by the government and guarded. One of its chief uses during the reign of Elizabeth I in the second half of the 16th century was as moulds for the manufacture of cannon balls. Graphite was transported from Keswick to London in armed stagecoaches. In 1751 an Act of Parliament was passed making it an offence to steal or receive “wad”. This crime was punishable by hard labour or transportation.

E

That the United States did not use pencils in the outer space till they spent \$1000 to make a pencil to use in zero gravity conditions is in fact a fiction. It is widely known that astronauts in Russia used grease pencils, which don't have breakage problems. But it is also a fact that their counterparts in the United States used pencils in the outer space before real zero-gravity pencil was invented. They preferred mechanical pencils, which produced fine lines, much clearer than the smudgy lines left by the grease pencils the Russians favoured. But the lead tips of these mechanical pencils broke often. That bit of graphite floating around the space capsule could get into someone's eye, or even find its way into machinery or electronics, causing an electrical short or other problems. But despite the fact that the Americans did invent zero-gravity pencils later, they stuck to mechanical pencils for many years.

F

Against the backcloth of a digitalized world, the prospect of pencils seems bleak. In reality, it does not. The application of pencils has by now become so widespread that they can be seen everywhere, such as classrooms, meeting rooms and art rooms, etc. A spectrum of users are likely to continue to use it into the future: students to do math works, artists to draw on sketch pads, waiters or waitresses to mark on order boards, make-up professional to apply to faces, and architects to produce blueprints. The possibilities seem limitless.

Questions 1–6

Complete the sentences below.

*Choose **ONE WORD ONLY** from the passage for each answer.*

Write your answers in boxes **1–6** on your answer sheet.

1. Graphite was found under a _____ in Borrowdale.
2. Ancient people used graphite to sign possession and number of _____.
3. The first pencil was graphite wrapped in _____ or animal skin.
4. In the eighteenth century, the _____ protected the mines when the value of graphite was realised.
5. During the reign of Elizabeth I, people were condemnable if they _____ or received the “wad”.
6. Russian astronauts preferred _____ pencils to write in outer space.

Questions 7–13

Do the following statements agree with the information? Write:

YES — if the statement agrees with the information

NO — if the statement contradicts the information

NOT GIVEN — if there is no information on this

7. Italy is probably the first country in the world to make pencils.
8. Germany used various kinds of wood to make pencils.
9. Graphite makes a pencil harder and sharper.
10. Pencils are not produced any more since the reign of Elizabeth I.
11. Pencil was used during the first American space expedition.
12. American astronauts did not replace mechanical pencils immediately after the zero-gravity pencils were invented.
13. Pencils are unlikely to be used in the future.

Test 2: The history of tea

The story of tea begins in China. According to legend, in 2737 BC, the Chinese emperor Shen Nung was sitting beneath a tree while his servant boiled drinking water, when some leaves from the tree blew into the water. Shen Nung, a renowned herbalist, decided to try the infusion that his servant had accidentally created. The tree was a Camellia sinensis, and the resulting drink was what we now call tea. It is impossible to know whether there is any truth in this story. But tea drinking certainly became established in China many centuries before it had even been heard of in the West. Containers for tea have been found in tombs dating from the Han Dynasty (206 BC—220 AD) but it was under the Tang Dynasty (618—906 AD), that tea became firmly established as the national drink of China.

It became such a favourite that during the late eighth century a writer called Lu Yu wrote the first book entirely about tea, the Ch'a Ching, or Tea Classic. It was shortly after this that tea was first introduced to Japan, by Japanese Buddhist monks who had travelled to China to study. Tea received almost instant imperial sponsorship and spread rapidly from the royal court and monasteries to the other sections of Japanese society.

So at this stage in the history of tea, Europe was rather lagging behind. In the latter half of the sixteenth century there are the first brief mentions of tea as a drink among Europeans. These are mostly from Portuguese who were living in the East as traders and missionaries. But although some of these individuals may have brought back samples of tea to their native country, it was not the Portuguese who were the first to ship back tea as a commercial import. This was done by the Dutch, who in the last years of the sixteenth century began to encroach on Portuguese trading routes in the East. By the turn of the century they had established a trading post on the island of Java, and it was via Java that in 1606 the first consignment of tea was shipped from China to Holland. Tea soon became a fashionable drink among the Dutch, and from there spread to other countries in continental western Europe, but because of its high price it remained a drink for the wealthy.

Britain, always a little suspicious of continental trends, had yet to become the nation of tea drinkers that it is today. Starting in 1600, the British East India Company had a monopoly on importing goods from outside Europe, and it is likely that sailors on these ships brought tea home as gifts. The first coffee house had been established in London in 1652, and tea was still somewhat unfamiliar to most readers, so it is fair to assume that the drink was still something of a curiosity. Gradually, it became a popular drink in coffee houses, which were as many locations for the transaction of business as they were for relaxation or pleasure. They were though the preserve of middle- and upper-class men; women drank tea in their own homes, and as yet tea was still too expensive to be widespread among the working classes. In part, its high price was due to a punitive system of taxation.

One **unforeseen** consequence of the taxation of tea was the growth of methods to avoid taxation—smuggling and adulteration. By the eighteenth century many Britons wanted to drink tea but could not afford the high prices, and their enthusiasm for the drink was matched by the enthusiasm of criminal gangs to smuggle it in. What began as a small time illegal trade, selling a few pounds of tea to personal contacts, developed by the late eighteenth century into an astonishing organised crime network, perhaps importing as much as 7 million lbs annually, compared to a legal import of 5 million lbs! Worse for the drinkers was that taxation also encouraged the adulteration of tea, particularly of smuggled tea which was not quality controlled through customs and excise. Leaves from other plants, or leaves which had already been brewed and then dried, were added to tea leaves. By 1784, the government realised that enough was enough, and that heavy taxation was creating more problems than it was words. The new Prime Minister, William Pitt the Younger, slashed the tax from 119 per cent to 12.5 per cent. Suddenly legal tea was affordable, and smuggling stopped virtually overnight.

Another great impetus to tea drinking resulted from the end of the East India Company's monopoly on trade with China, in 1834. Before that date, China was the country of origin of the vast majority of the tea imported to Britain, but the end of its monopoly stimulated the East India Company to consider growing tea outside China. India had always been the centre of the Company's operations, which led to the increased cultivation of tea in India, beginning in Assam. There were a few false starts, including the destruction by cattle of one of the earliest tea nurseries, but by 1888 British tea imports from India were for the first time greater than those from China.

The end of the East India Company's monopoly on trade with China also had another result, which was more dramatic though less important in the long term: it ushered in the era of the tea clippers. While the Company had had the monopoly on trade, there was no rush to bring the tea from China to Britain, but after 1834 the tea trade became a virtual free for all. Individual merchants and sea captains with their own ships raced to bring home the tea and make the most money, using fast new clippers which had sleek lines, tall masts and huge sails. In particular there was a competition between British and American merchants, leading to the famous clipper races of the 1860s. But these races soon came to an end with the opening of the Suez Canal, which made the trade routes to China viable for steamships for the first time.

Questions 1-7

Complete the sentences below with words taken from Reading Passage 1

Use **ONE WORD** for each answer. @Teaxona

Write your answers in boxes **1-7** on your answer sheet.

- 1 Researchers believed the tea containers detected in from the Han Dynasty was the first evidence of the use of tea.
- 2 Lu Yu wrote a about tea before anyone else in the eighth century.
- 3 It was from Japan who brought tea to their native country from China.
- 4 Tea was carried from China to Europe actually by the
- 5 The British government had to cut down the taxation on tea due to the serious crime of
- 6 Tea was planted in besides China in the 19th century.
- 7 In order to compete in shipping speed, traders used for the race.

Questions 8 – 13

Do the following statements agree with the information given in Reading Passage 1?

In boxes **8-13** on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this

- 8 Tea was popular in Britain in the 16th century.
- 9 Tea was more fashionable than coffee in Europe in the late 16th century.
- 10 Tea was enjoyed by all classes in Britain in the seventeenth century.
- 11 The adulteration of tea also prompted William Pitt the Younger to reduce the tax.
- 12 Initial problems occurred when tea was planted outside China by the East India Company.
- 13 The fastest vessels were owned by America during the 19th century clipper races.

Test 3: A Brief History of Humans and Food

During the journey from our hunter-gatherer ancestors to the present day there have been three seismic changes that have had an impact on the food we eat: the discovery of cooking, the emergence of agriculture and the invention of methods of preserving food.

The 19th-century scientist Charles Darwin thought that cooking, after language, was the greatest discovery made by man. All of us eat some raw food, for instance fruit and vegetables, but the great majority of food we eat is cooked. Cooking can turn plants that are inedible into edible food by destroying toxic chemicals that plants often manufacture to protect themselves against attack by insects or other herbivorous animals. These toxic chemicals are referred to as 'plant secondary compounds', because they are not directly involved in the plant's normal growth, development and reproduction, and are produced purely as chemical defences. They give many of the plants we consume, such as coffee or brussels sprouts, their bitter taste.

Cooked food is often more digestible, because heat breaks down tough cellulose cell walls in plants or tough connective tissue in animals. Chewing raw turnip, a plate of uncooked rice, or a raw leg of lamb is much harder work than eating the cooked equivalent. The energy expended in chewing to break down the tough material is replaced by energy from the fuel that is used in cooking the food, so the ratio of energy gained to energy expended by the body is greater when food is cooked.

Until the development of agriculture, hunter-gatherers spent up to seven hours a day gathering food. This all began to change around 10,500 years ago with the advent of farming, which led to some dramatic changes in human societies. People began to create a variety of new tools to help with survival, and in turn populations increased in size. These changes led to the possibility of specialisation of different tasks within society. It was around this time that writing became more sophisticated and allowed people to maintain records of the harvest and taxes. Eventually, formalised structures of government were established as people settled in one area.

The arrival of agriculture meant that, for the first time, our ancestors had more food than they could eat immediately. This, combined with the seasonality of production, led them to discover methods of preserving food: smoking, drying, adding acid by fermentation or adding salt. These four methods all share one feature in common: they make the food a more

hostile environment for bacteria that can cause it to spoil. They also tend to slow down any natural chemical reactions in the food that would cause decay.

Although foods today are still preserved in the ancient ways, two more recent methods of preserving food have become more common: canning and freezing. Canning was invented by a Frenchman, Nicholas Appert, in the early 19th century. He sealed food in bottles fabricated from glass and then heated them in boiling water to cook the contents. Appert's method had great advantages over older methods of food preservation: it could be applied to a wide range of foods, and the flavour of the food as well as the texture were similar to the freshly cooked product. His idea was soon copied by an Englishman, Peter Durand. Until this point containers had weighed too much to be widely used, but he produced the first ones which were light and resistant to damage. Two years later, in 1812, two Englishmen, Bryan Donkin and John Hall, started the commercial canning of food, although the real take-off in popularity of canning had to wait until the can opener was invented in 1855. Up to this time, cans were opened with a chisel which was used to break open the top when hit with a hammer. Canning is an extremely effective way of preserving food: one can which contained meat dating from 1824 was opened in 1939 and the contents were still in good condition.

In the 21st century, the dominance of canning as a method of food preservation has been overtaken by another technology: freezing. Chilling food to keep it fresh is an old idea. The earliest mentions of icehouses, thick-walled buildings, half underground, date back to 1700 BC in northwest Iran. In early 16th-century Italy, water was mixed with chemicals to lower its freezing point to -18 degrees Celsius, and several centuries later frozen fish and other goods were transported by ship from Australia to England. But the modern frozen food industry was started in the 1920s by an American, Clarence Birdseye. While Birdseye was on a fishing trip with the Inuit in the Canadian Arctic, he observed that very rapid freezing creates smaller ice crystals and therefore causes less damage to food. This was something he had not expected. Nevertheless, the big growth in demand for frozen food came about with the arrival of freezers in the homes of ordinary people. The advantages of frozen over canned food include the fact that the flavour and consistency are often identical to the equivalent fresh product, and that freezing can be used to preserve a huge variety of foods.

Questions 1–5

Do the following statements agree with the information given in Reading Passage 1?

In boxes **1–5** on your answer sheet, write:

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this @Teaxona

- 1** According to Darwin, cooking was the most significant development in human history.
- 2** The process of cooking gets rid of some plant poisons.
- 3** Eating cooked food is more energy efficient than eating raw food.
- 4** Clarence Birdseye had previously worked in the Australian food industry.
- 5** Birdseye's trip with the Inuit confirmed what he already believed about rapid freezing.

Questions 6–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes **6–13** on your answer sheet.

The changes agriculture brought about were:

6 the development of equipment and larger _____

7 the ability to keep _____ as writing developed

8 early methods included smoking, drying and adding acid or _____

Canning

9 Appert put food into containers made of _____

10 preserved food had the same taste and _____ as fresh food

11 Durand produced cans which were _____ and hard to break

12 before can openers, cans were opened with a _____

13 food remained edible in a can of _____ from 1824

Test 4: The History of The Poster

The appearance of the poster has changed continuously over the past two centuries. The earliest posters were known as *broadsides* and were used for public and commercial announcements. Printed on one side using metal type, they were produced quickly and crudely in large quantities. Since they were designed to be read from a distance, they required very large lettering.

However, large metal type had several disadvantages: it was expensive, took up a great deal of storage space and was extremely heavy. Even if a printer owned such type, there were rarely enough letters available, resulting in posters that mixed and matched styles.

Commercial demand for large type was met with the development of wood type production. In 1827, Darius Wells invented a special wood drill – the lateral router – capable of cutting letters into wooden blocks. Used alongside William Leavenworth's pantograph (1834), it enabled craftsmen to create decorative wooden letters of various shapes and sizes. Early posters produced in this way lacked colour and coherent design, and typically combined wooden and metal type in a mixture of styles.

A major step forward came with the application of lithography, invented by Alois Senefelder in 1796, which enabled artists to draw letters by hand and opened the door to unlimited stylistic possibilities. The method involved drawing with a greasy crayon on smooth limestone and transferring the image to paper. Although this process accurately captured the artist's intention, the final printed image appeared in reverse, so letters and images had to be drawn backwards, often using a mirror or transfer paper.

This difficulty limited the impact of lithography on posters until the 1860s, when Jules Chéret developed his *three-stone lithographic process*. This technique offered artists the chance to experiment with a wide spectrum of colours. Despite its complexity, it produced striking results, with subtle tones impossible in other media even today. The ability to combine words and images in an attractive and economical format finally established the lithographic poster as a powerful new art form.

From the 1870s onwards, posters became the chief medium for advertising before the rise of magazines, and the dominant form of mass communication in the rapidly growing cities of Europe and America. In Paris, Milan and Berlin, posters became so popular that they were stolen from the streets almost as soon as they were put up. Chéret, later known as "the father of the modern poster", organised the first major poster exhibition in 1884 and published the first book on poster art two years later. He also arranged for artists to produce smaller versions suitable for display in people's homes.

During the 1890s, posters spread widely and came to celebrate each region's cultural institutions: cafés in France, opera and fashion in Italy, festivals in Spain, literature in Holland and trade fairs in Germany. Poster exhibitions were held in Britain and Italy in 1894, in Germany in 1896, and in Russia in 1897. To many observers, the most significant of all was in Reims, France, in 1896, displaying 1,690 posters arranged by country.

In the early 20th century, the poster continued to be a major communication tool, evolving through an array of styles. By the 1950s, however, it shared public attention with radio and print. Most posters by then were produced using photo offset, the mass-production technique that creates the familiar dot pattern seen in newspapers and magazines. Photography, first used in Russian posters in the 1920s, became increasingly common.

In the late 1950s, a new graphic trend emerged, relying heavily on black-and-white typographic elements. Known as the International Typographic Style, it used a mathematical grid, strict design rules and monochrome photography to create a clear and logical appearance. It became the leading global style in the 1970s and still influences graphic design today.

Its clarity made it ideal for the post-war international marketplace, where accessibility of text and symbols was essential. Corporations sought internationally recognisable identities, and worldwide events such as the Olympics required universal visual solutions, both of which the new style provided.

By the late 1970s, however, the International Typographic Style was criticised for being cold, formal and overly rigid. In response, Wolfgang Weingart, inspired by his colleagues, experimented with offset printing to produce posters that appeared chaotic, playful and spontaneous – a dramatic departure from previous conventions. His innovative approach laid the groundwork for several new movements, including Memphis, Retro, and the early developments in computer-based graphic design.

Questions 1–5

Complete the table below.

Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

Write your answers in boxes **1–5** on your answer sheet.

Early Printing Methods

Method	Features	Problems
Metal type	• produced large print	• cost, weight and 1 _____ • mixed styles
Wood type	• wood drill used with 2 _____ • produced a range of letters	• lacked both 3 _____ and organized design • mixed type @Teaxona
Lithography	• letters drawn by hand • design tool: a 4 _____	• had to use a mirror or 5 _____ to achieve the correct image

Questions 6–9

Complete the flow chart below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Write your answers in boxes **6–9** on your answer sheet.

Jules Chéret

- 1860s** – invention of *three-stone lithographic process*
 combination of both **6** _____ and illustrations on coloured posters
- 1870s** – posters used for advertising and **7** _____ in Europe
- 1884–86** – Chéret's poster **8** _____ and book on poster art
- 1890s** – posters representing **9** _____ around the world

Questions 10–13 @Teaxona

Do the following statements agree with the information given in the passage?
In boxes **10–13** on your answer sheet, write:

TRUE if the statement agrees
FALSE if it contradicts the information
NOT GIVEN if there is no information

- 10.** By the 1950s, photographs were more common than drawings on posters.
- 11.** Characteristics of the Typographic Style can still be seen in modern posters.
- 12.** The Typographic Style fulfilled an international need at a particular time.
- 13.** Weingart developed his ideas based on suggestions from his students.

Test 5: The History of Glass

From our earliest origins, man has been making use of glass. Historians have discovered that a type of natural glass – obsidian – formed in places such as the mouth of a volcano as a result of the intense heat of an eruption melting sand – was first used as tips for spears. Archaeologists have even found evidence of man-made glass which dates back to 4000 BC; this took the form of glazes used for coating stone beads. It was not until 1500 BC, however, that the first hollow glass container was made by covering a sand core with a layer of molten glass.

Glass blowing became the most common way to make glass containers from the first century BC. The glass made during this time was highly coloured due to the impurities of the raw material. In the first century AD, methods of creating colourless glass were developed, which was then tinted by the addition of colouring materials. The secret of glass making was taken across Europe by the Romans during this century. However, they guarded the skills and technology required to make glass very closely, and it was not until their empire collapsed in 476 AD that glass-making knowledge became widespread throughout Europe and the Middle East. From the 10th century onwards, the Venetians gained a reputation for technical skill and artistic ability in the making of glass bottles, and many of the city's craftsmen left Italy to set up glassworks throughout Europe.

A major milestone in the history of glass occurred with the invention of lead crystal glass by the English glass manufacturer George Ravenscroft (1632–1683). He attempted to counter the effect of clouding that sometimes occurred in blown glass by introducing lead to the raw materials used in the process. The new glass he created was softer and easier to decorate, and had a higher refractive index, adding to its brilliance and beauty, and it proved invaluable to the optical industry. It is thanks to Ravenscroft's invention that optical lenses, astronomical telescopes, microscopes and the like became possible.

In Britain, the modern glass industry only really started to develop after the repeal of the Excise Act in 1845. Before that time, heavy taxes had been placed on the amount of glass melted in a glasshouse, and were levied continuously from 1745 to 1845. Joseph Paxton's Crystal Palace at London's Great Exhibition of 1851 marked the beginning of glass as a material used in the building industry. This revolutionary new building encouraged the use of glass in public, domestic and horticultural architecture. Glass manufacturing techniques also improved with the advancement of science and the development of better technology.

From 1887 onwards, glass making developed from traditional mouth-blowing to a semi-automatic process, after factory-owner H. M. Ashley introduced a machine capable of producing 200 bottles per hour in Castleford, Yorkshire, England – more than three times quicker than any previous production method. Then in 1907, the first fully automated machine was developed in the USA by Michael Owens –

founder of the Owens Bottle Machine Company (later the major manufacturers Owens-Illinois) – and installed in its factory. Owens' invention could produce an impressive 2,500 bottles per hour. Other developments followed rapidly, but it was not until the First World War, when Britain became cut off from essential glass suppliers, that glass became part of the scientific sector. Previously, glass had been seen as a craft rather than a precise science.

Today, glass making is big business. It has become a modern, hi-tech industry operating in a fiercely competitive global market where quality, design and service levels are critical to maintaining market share. Modern glass plants are capable of making millions of glass containers a day in many different colours, with green, brown and clear remaining the most popular. Few of us can imagine modern life without glass. It features in almost every aspect of our lives – in our homes, our cars and whenever we sit down to eat or drink. Glass packaging is used for many products; many beverages are sold in glass, as are numerous foodstuffs, as well as medicines and cosmetics.

Glass is an ideal material for recycling, and with growing consumer concern for green issues, glass bottles and jars are becoming ever more popular. Glass recycling is good news for the environment. It saves used glass containers being sent to landfill. As less energy is needed to melt recycled glass than to melt down raw materials, this also saves fuel and production costs. Recycling also reduces the need for raw materials to be quarried, thus saving precious resources.

Questions 1–8

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes **1–8** on your answer sheet.

The History of Glass

- Early humans used a material called **1** _____ to make the sharp points of their **2** _____.
- 4000 BC: **3** _____ made of stone were covered in a coating of man-made glass. @Teaxona
- First century BC: glass was coloured because of the **4** _____ in the material.
- Until 476 AD: only the **5** _____ knew how to make glass.
- 17th century: George Ravenscroft used **6** _____ to prevent **7** _____ in blown glass.
- Mid-19th century: British glass production developed after changes to laws concerning **8** _____.

Questions 9–13

Do the following statements agree with the information in the passage?

TRUE — *if the statement agrees with the information*

FALSE — *if the passage contradicts the statement*

NOT GIVEN — *if the passage does not say @Teaxona*

9. In 1887, H. M. Ashley had the fastest bottle-producing machine that existed at the time.
10. Michael Owens was hired by a large US company to design a fully automated bottle-manufacturing machine for them.
11. Nowadays, most glass is produced by large international manufacturers.
12. Concern for the environment is leading to an increased demand for glass containers.
13. It is more expensive to produce recycled glass than to manufacture new glass.

Test 6: The History of The Picnic

Picnics became popular in France in the 18th century. A favorite pastime of the upper classes, they were initially purely indoor affairs, held at home or in hired rooms. It was assumed that invited guests would contribute – they could either bring a dish or pay a share of the overall cost. These picnics were often marked by conversation and wit and, as such, were associated with intellectual refinement. At larger gatherings, there was also music, with the result that 'picnic' came to mean the same as a ball or party. Picnics were, however, nothing if not adaptable. When the first restaurants appeared, picnickers quickly began going to them and having their picnics there.

The French Revolution in 1789 changed everything, with many wealthy French aristocrats fleeing abroad. Some went to Austria, others to Prussia or even America, but the majority chose to go to England. Settling mainly in London, some were now in need of money, but they did their best to maintain their old way of life – and in doing so, introduced the picnic to England. This led to several important developments. The first, and shortest-lived, was that the picnic became decidedly less refined and noisier. This was thanks to a group of around 200 wealthy young people who loved France and French culture and, in late 1801, founded the 'Pic Nic Society'.

Held in hired rooms on Tottenham Street in London, the society's gatherings were intended to impress. Every member was required to bring a costly dish and something to drink. As The Times newspaper reported, each member tried to outdo the others in luxury and expense. After dinner, the principal entertainment was always a play. The productions were, admittedly, quite basic: the stage was small and crudely built, unlike in a theatre, and the actors were never professionals. But even so, they were wildly popular.

Another development in England in the 1800s was more profound. Picnics were adopted by the emerging middle classes and moved outside. What caused this change is unclear, but the most likely explanation is that the English simply applied a fashionable French word to a pre-existing practice, without being aware of what it meant. One of the results was that music ceased to be provided at picnics, which became just rather plain, open-air meals to which people were invited by a host. Another was that it became both more 'genteel' and – thanks to the idealization of the countryside – more innocent.

In the decades which followed, this form of picnic found its way to the United States. As was only to be expected, it remained a refined activity most popular with the middle classes in cities; but unlike in England, its setting was associated more with a flight from civilization than with childlike simplicity. Though still tinged with innocence, paintings of picnics by well-known American artists – such as those by Winslow Homer – tended, therefore, to contain fewer gently rolling fields than shown by English artists and instead had more gnarled trees and rough mountains.

Picnics in the open air were less well received elsewhere, however. Though the restoration of the French monarchy was accompanied by the return of picnicking to France, their indoor version still predominated, especially among the aristocracy. But the legacy of revolutionary equality in France also opened up the picnic there to others lower down the social scale. By the mid-19th century, even members of the working classes in France were starting to adopt the practice. When outdoor picnics eventually did begin to gain ground in France, they were regarded with considered suspicion. Perhaps because of a growing reaction in France against the Romantic idealization of nature, they were not seen as innocent and wholesome, as they were in England and the United States.

In general, not until the early 20th century did the outdoor picnic prevail over the indoor kind. Particularly in England, the development of different forms of transport and the acceleration of social change made the countryside accessible to a far greater proportion of the population. Before long, the popularity of the picnic had grown so much that specialist picnic baskets were being produced for the mass market and their contents were becoming more standardized.

Over the last 100 years, picnics have undergone still further changes – largely as a result of the international availability of a huge variety of food to be transported over long distances. Picnics will undoubtedly continue to evolve in the future. Shifting patterns of trade will almost certainly continue to change the foods we pack for a picnic. But whatever happens, one thing is certain: as long as there are friends with whom to share a meal, there will be few things so pleasant as a picnic lunch.

Questions 1–7

Do the following statements agree with the information given in Reading Passage 1
In boxes 1-7 on your answer sheet, write

TRUE	<i>if the statement agrees with the information</i>
FALSE	<i>if the statement contradicts the information</i>
NOT GIVEN	<i>if there is no information on this @Teaxona</i>

- 1 The first picnics in 18th-century France were held in parks.
- 2 People who attended early French picnics were expected to bring food or help with the expense.
- 3 The hosts of early French picnics often asked their guests to play music.
- 4 The picnic was brought to England by aristocrats escaping the French Revolution.
- 5 Many complaints were received about the noise at the first English picnics.
- 6 People who wanted to join the Pic Nic Society had to have the approval of existing members.
- 7 The picnics held by the Pic Nic Society were extravagant events.

Questions 8–13

Complete the notes below. @Teaxona

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 9–16 on your answer sheet.

Changes in picnics

In England

- in the 1800s, picnics began to be held **8**
- picnics no longer usually included **9**.....for entertainment.

In the USA

- picnics were a favorite activity for people living in **10**
- American paintings of picnics were different from those by English artists

Return to France

- indoor picnics were still preferred by members of the **11**

20th century onwards

- new kinds of **12**..... meant that picnics in the countryside became more common
- **13**.....designed especially for use by picnickers were widely sold

Test 7: The History of Bicycles

The bicycle was not invented by one individual or in one country. It took nearly 100 years and many individuals for the modern bicycle to be born. By the end of those 100 years, bicycles had revolutionized the way people travel from place to place.

Bicycles first appeared in Scotland in the early 1800s, and were called velocipedes. These early bicycles had two wheels, but they had no pedals. The rider sat on a pillow and walked his feet along the ground to move his velocipede forward.

Soon a French inventor added pedals to the front wheel. Instead of walking their vehicles, riders used their feet to run the pedals. However, pedaling was hard because velocipedes were very heavy. The framework was made of solid steel tubes and the wooden wheels were covered with steel. Even so, velocipedes were popular among rich young men, who raced them in Paris parks.

Because of the velocipedes were so hard to ride, no one thought about using them for transportation. People didn't ride velocipedes to the market or to their jobs. Instead, people thought velocipedes were just toys.

Around 1870, American manufacturers saw that velocipedes were very popular overseas. They began building velocipedes, too, but with one difference. They made the frameworks from hollow steel tubes. This alteration made velocipedes much lighter, but riders still had to work hard to pedal just a short distance. In addition, roads were bumpy so steering was difficult. In fact, most riders preferred indoor tracks where they could rent a velocipede for a small fee and take riding lessons.

Subsequent changes by British engineers altered the wheels to make pedaling more efficient. They saw that when a rider turned the pedals once, the front wheel turned once. If the front wheel was small, the bicycle traveled just a small distance with each turn. They reasoned that if the front wheel were larger, the bicycle would travel a greater distance. So they designed a bicycle with a giant front wheel. They made the rear wheel small. Its primary purpose was to help the rider balance. Balancing was hard because the rider had to sit high above the giant front wheel in order to reach the pedals.

This meant he was in danger of falling off the bicycle and injuring himself if he lost his balance. Despite this inherent danger, "high wheelers" became very popular in England.

American manufacturers once again tried to design a better bicycle. Their goal was to make a safer bicycle. They substituted a small wheel for the giant front wheel and put the driving mechanism in a larger rear wheel. It would be impossible for a rider to pedal the rear wheel, so engineers designed a system of foot levers. By pressing first the right one and then the left, the rider moved a long metal bar up and down. This bar turned the rear axle. This axle turned the rear wheel and the bicycle minimized the dangers

inherent in bicycle riding, more and more people began using bicycles in their daily activities.

The British altered the design one last time. They made the two wheels equal in size and created a mechanism that uses a chain to turn the rear wheel. With this final change, the modern bicycle was born.

Subsequent improvements, such as brakes, rubber tires, and lights were added to make bicycles more comfortable to ride. By 1900, bicycle riding had become very popular with men and women of all ages. Bicycles revolutionized the way people worldwide ride bicycles for transportation, enjoyment, sport, and exercise.

Questions 1–6

Complete the sentences below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

In the invention of the bicycle took part not only one individual or the country, but the world in general during the **1**..... years.

They had two wheels, but without **2**.....

They were regarded as **3**.....

Manufacturers in America found that this invention is wide spread **4**.....

Frameworks were made from **5**..... which were empty inside.

The British inventors concluded that if the front wheel were large, the bicycle would travel a **6**..... distance.

American manufacturers changed the wheel structure with the aim of creating a **7**.....

The last alteration was using the **8**..... to spin the rear wheel.

Eventually, bicycles were upgraded with stopping systems, lights and tires made of **9**.....

Questions 7–10

Do the following statements agree with the information in Reading Passage 1? Write:

TRUE if the statement agrees with the passage

FALSE if it contradicts the passage

NOT GIVEN if there is no information @Teaxona

10. The bicycle was invented by Americans only.
11. The weight was one of the issues connected with earlier models of bikes.
12. The French inventor made the pedal from wood.
13. Keeping balance in early British bicycles was quite straightforward.

Test 8: The History of the Guitar

A The earliest stringed instruments currently known to archaeologists are bowl harps. For millennia, people made bowl harps using, for example, tortoise shells as resonators, with a bent stick for a neck and one or more gut or silk strings. The world's museums contain many such harps from the ancient Sumerian, Babylonian, and Egyptian civilisations. Around 2500–2000 BC more advanced harps, such as the beautifully carved 11-stringed instrument found in the tomb of Queen Shub-Ad in ancient Mesopotamia, now modern-day Iraq, started to appear.

B The tanbur probably developed from the bowl harp. It was different from the bowl harp in that its neck was straightened out to allow the strings to be pressed down to create more notes. Tomb paintings and stone carvings in Egypt indicate that harps and tanburs — plus flutes and percussion instruments — were being played together 3,500–4,000 years ago. Archaeologists have also found many similar relics amongst the ruins of the ancient Mesopotamian civilisation. Many of these instruments have survived into modern times in almost unchanged form, for example folk instruments of the region such as the Turkish saz and Afghan panchtar.

C At 3,500 years old, the tanbur which belonged to the Egyptian singer Har-Mose is the earliest known example of this instrument. Har-Mose's tanbur had three strings and a plectrum suspended from the neck by a cord. The soundbox, which increased the volume, was made of beautifully polished cedarwood and covered in rawhide. It can be seen today at the Archaeological Museum in Cairo.

D In order to distinguish guitars from other stringed instruments, it is helpful to have a broad definition of the guitar. Music expert Dr Michael Kasha defines a guitar as having “a long, fretted neck, flat wooden soundboard, ribs, and a flat back, most often with sides that curve inwards.” The oldest known visual representation of such an instrument is a stone carving at Alaca Huyuk in Turkey, which shows a 3,300-year-old instrument with a long neck and sides that clearly curve inwards.

E The name “guitar” comes from the ancient Sanskrit word for “string” — *tar*. Many popular stringed instruments used in central Asia today have existed in an unchanged form for several thousand years, as shown by archaeological finds in the area. Many have names that end in *tar*, with a prefix indicating the number of strings, such as the *doter*, a two-stringed instrument found in Turkestan, and the Persian three-stringed *setar* and four-stringed *chartar*. The Indian *sitar* almost certainly took its name from the *setar*, but over the centuries it evolved radically, following the Indians' own aesthetic and cultural ideals.

F Tanburs and harps spread around the ancient world with travellers, merchants and seamen. The earliest guitar-like instruments to arrive in Europe had, most often, four strings. Many such instruments, and variations with from three to five strings, can be seen in mediaeval illustrated manuscripts. They were also carved in stone in European churches and cathedrals, from the first century AD through until the 13th century.

G When the four-stringed Persian *chartar* arrived in Spain, however, it changed in form and construction, acquiring pairs of strings tuned to the same note instead of single strings. It became known as the *chitarra*. By the middle of the 14th century, the chitarra had become dominant, at least in most of Europe. The earliest known music for the eight-stringed chitarra was written in 16th-century Spain. The ten-string version first appeared in Italy at the same time, and gradually replaced the eight-stringed instrument. A further two strings first appeared in the 17th century, an innovation which guitar makers all over Europe quickly took up. However, this twelve-string arrangement gradually gave way to six single strings across the continent. The six-stringed guitar can thus be said to be a development of the twelve-stringed, rather than vice versa, as was thought previously.

H At the beginning of the 19th century, the present-day guitar began to take shape, although bodies were still fairly small and narrow-waisted. The modern classical guitar first appeared in its current form in the mid-19th century, when the Spanish guitar maker Antonio Torres increased the size of the body, altered its proportions, and introduced the revolutionary fan-braced top. His design radically increased the volume and improved the tone of the instrument, and very soon became the norm. This design has remained essentially unchanged to this day.

I At the time when Torres made his breakthrough, German immigrants to America — among them Christian Friedrich Martin — began making guitars with X-braced tops. Steel strings, which became widely available several decades later in the early 1900s, offered the promise of a much louder guitar, but the increased tension was too much for the fan-braced top. The stronger X-braced top proved equal to the job, and quickly became the industry standard.

J At the end of the 19th century, guitar manufacturer Orville Gibson added steel strings to a body constructed like a cello, a combination which produced more volume. The electric guitar was born when pickups were fitted to Hawaiian and jazz guitars in the late 1920s, but met with little success until 1936, when Gibson introduced its famous ES150 model.

Questions 1–6 @Teaxona

Do the following statements agree with the information given in Reading Passage 1?

In boxes **1–6** on your answer sheet, write:

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information`

NOT GIVEN if there is no information on this @Teaxona

1. The instrument found in Queen Shub-Ad's tomb is the world's oldest known version of a harp.
2. Today's Afghan panchtar is very similar to an ancient Mesopotamian instrument.
3. The Egyptian singer Har-Mose was an excellent tanbur player.
4. Cairo Archaeological Museum contains many historic musical instruments.
5. The instrument carved in stone at Alaca Huyuk is consistent with Dr Michael Kasha's definition of a guitar.
6. The different instruments that appeared in medieval literature had the same number of strings.

Questions 7–13

Complete the table below.

Choose **ONE WORD ONLY** from the passage for each answer.

The Development of the Guitar

Date	Type of instrument	Notes
13th–19th century	Chitarra @Teaxona	– was a development of an earlier instrument called the 7. _____ – extra strings added in 16th century in Italy
from around the 1850s	Classical guitar	– its shape a result of modifications including a larger 8. _____ introduced by Antonio Torres
		– changes produced better tone and greater 9. _____
mid-19th century	X-braced top guitar	– first made in 10. _____
around 1900		– strings made of 11. _____ became available
1920s onwards	Electric guitar	– in the 1920s, 12. _____ added to guitars
		– a well-known version brought out by 13. _____

Test 9: The History of the Chicken

The story begins 10,000 years ago in a jungle in Asia and ends today in kitchens all over the world.

Chicken as the universal food of our era, crossing cultural boundaries with ease. With its mild taste and uniform texture, chicken presents a blank canvas for the national flavours of almost any cuisine. However, until the introduction of large-scale industrial production in the 20th century, the economic and nutritional contribution of chickens was small. Chickens were considered as domestic birds that were useful to humanity, but unlike other animals such as the horse or the ox, they did little to change the course of history. Nonetheless, the chicken has inspired contributions to culture, art, cuisine, science and religion over the millennia. Chickens were, and still are, a sacred animal in some cultures. The hen is a worldwide symbol of nurturing and fertility. The rooster is a universal sign of masculinity.

The domesticated chicken has a complicated family tree, stretching back 10,000 years. Recent DNA testing has shown that the chicken's wild ancestor is the red junglefowl. The bird's resemblance to modern chickens is seen in the red feathers on top of its head, the red flesh around its mouth, and its distinctive call in the morning. The females of this breed protect their eggs in the same way modern-day chickens do, and they make the same clucking sounds. In its habitat, which stretches from northeastern India to the Philippines, the red junglefowl looks on the forest floor for insects, seeds and fruit and flies up to nest in the trees at night. That's about as much flying as it can manage, a trait that made it relatively easy for humans to capture and domesticate it. But the red junglefowl is not the sole ancestor of the modern chicken. Scientists have identified three closely related species that may have bred with the red junglefowl. Precisely how much genetic material these other birds contributed to the DNA of domesticated chickens is unknown. However, recent research suggests that modern chickens inherited at least one trait, their yellow skin, from the grey junglefowl of southern India.

Once chickens were domesticated, cultural contacts, trade, migration and territorial conquest resulted in their introduction in different regions around the world over several thousand years. Chickens arrived in Egypt in around 1750 BC, as fighting birds and additions to zoos. Drawings of the birds decorated royal tombs, with cockfighting scenes that involved betting matches and offerings to the afterlife. It would take another 100 years before the bird became popular among ordinary Egyptians.

It was in that era that Egyptians mastered the technique of artificial incubation, which freed hens to lay more eggs. This was not easy. Most chicken eggs will hatch in three weeks, but only if the temperature remains at around 38 degrees Celsius. The eggs must also be turned three to five times a day, and the humidity must remain stable at about 55 percent, otherwise physical deformities can result, or the eggs won't hatch. The

Egyptians constructed vast complexes of large rooms, which were essentially ovens, and these were connected to a series of corridors and vents that allowed attendants to regulate the heat from fires. The egg attendants kept their methods a secret from outsiders for centuries.

Around the Mediterranean, archaeologists have uncovered chicken bones from about 800 BC. Chickens were a delicacy among the Romans, whose culinary innovations included the...

omelette, and the practice of stuffing birds for cooking. European farmers began developing methods to fatten the birds, such as feeding them bread which had been soaked in wine. But the chicken's status in Europe appears to have diminished with the collapse of Rome. In the period after the fall of the Roman Empire, chicken farms vanished and the birds returned to the size they had been 1,000 years earlier.

Well into the 20th century chickens, although valued as a source of eggs, played a relatively minor role in diet and the economy. Long after cattle had entered the industrial age of centralized and mechanized large-scale production, chicken production was still mostly a small-scale local enterprise. The breakthrough that made today's huge bird farms possible was the inclusion of antibiotics, along with a mixture of different vitamins, in the food chickens ate. This allowed chickens to be raised indoors and be protected from cold temperatures and heavy rain as well as predators. This factory farming represents the chicken's final step in its transformation into a big protein-producing business.

Factory farms turning out an increasing amount of chicken have resulted in an increasing demand. Modern chickens are giants in a system designed to convert grain into protein with staggering efficiency. It takes less than 1 kilogram of feed to produce 400 grams of chicken. By comparison, around 3 kilograms of feed are required to produce 400 grams of beef. A day-old chick can be turned into a 2 kilogram bird in six weeks – half the time it took two generations ago.

Chickens also make wonderful pets, as breeders will tell you. They are as colourful as tropical fish but more affectionate, as cute as guinea pigs and better at catching mice than cats. In the USA, exotic and heritage breeds of chicken are being sold for considerable sums of money as the fashion for keeping chickens in the backyard becomes more popular.

All in all, the globe-spanning chicken is an epic story of evolutionary, agricultural and culinary success, outnumbering human beings on the planet by nearly three to one.

Questions 1–7 @teaxona

Do the following statements agree with the information given in Reading Passage 1?

In boxes **1–7** on your answer sheet, write:

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this @Teaxona

1. Chicken is globally popular because it can be used for different styles of cooking.
2. Chickens have poor capacity for flight compared to junglefowl.
3. Scientists believe that the domestic chicken has more than one ancestor.
4. A modern chicken has the same skin colour as the grey junglefowl.
5. After the Roman Empire ended, chicken's popularity in Europe declined.
6. Some people criticise the methods involved in factory farming chickens.
7. In the USA, fewer people are keeping expensive breeds of chicken at home.

Questions 8–13

Complete the sentences below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Write your answers in boxes **8–13** on your answer sheet. @Teaxona

The domestication and farming of chickens

Egypt

- In 1750 BC the first chickens were seen in Egyptian 8. _____ and used in fights.
- Pictures of chickens were used on the 9. _____ of kings and queens.
- Egyptians developed a method of egg 10. _____ that didn't involve hens.
- For eggs to hatch, heat and 11. _____ levels had to stay the same.
- The eggs were hatched in rooms that functioned as 12. _____.

Romans

- Farmers gave chickens a mixture of bread and wine to make them bigger.

20th-century farming

- Farmers started adding antibiotics and vitamins to chicken feed.
- Chickens were kept inside, safe from bad weather and 13. _____.

Test 10: The History of Cakes at Weddings

In Western cultures, since antiquity, weddings customarily have been celebrated with a special cake. Ancient Roman wedding ceremonies were finalized by breaking a cake of wheat or barley over the bride's head as a symbol of good fortune. The newly married couple then ate some of the crumbs together. Afterward, the wedding guests gathered up the remaining crumbs as tokens of good luck. Wedding guests were also supplied with sweetmeats called confetti, a sweet mixture of nuts, dried fruit and honeyed almonds. Handfuls of confetti were showered over the bride and groom; indeed, it seems to have been the custom to throw confetti enthusiastically. Eventually, confetti in the form of sweets and nuts was replaced with rice, flower petals, or colored paper, and these new types of confetti continue to be thrown over newly married couples in many countries around the world.

When the Romans invaded Britain in 43 AD, many of their customs and traditions became part of British life. These customs included their wedding customs, and when the Normans invaded Britain in 1066 they brought many French traditions. Other changes came about due to increased trade and contact with Europe, but present-day British wedding traditions remain firmly rooted in the past. In medieval Britain, small spiced buns, which were common everyday fare, were often eaten at weddings. These were stacked in a towering pile, as high as possible. If the bride and groom were able to kiss over the tall stack, it augured a lifetime of riches. The earliest British recipe exclusively for eating at weddings is Bride Pie, which was recorded in 1685. This was a large round pie with an elaborately decorated pastry crust that concealed a filling of oysters, pine nuts, lamb and spices. Each guest had to eat a small piece of the pie or not to do so was considered extremely impolite. A ring was traditionally placed in the pie, and the lady who found it would be next to marry.

In the 17th century, Bride Pie was changed into Bride Cake, the predecessor of the modern British centrepieces for weddings. Some people made Bride Cake in the cheaper form of two large rounds of pastry sandwiched together with currants and sprinkled with sugar. Very few homes at the time could boast of having ovens, but this type of pastry cake could be cooked on a baking stone on the hearth.

Later in the 17th century, there was a new development when wedding cakes began to be made in pairs, one for the bride and another for the groom. Both cakes were dark, heavy fruitcakes; the groom's cake was smaller than the bride's cake, and was cut up into little squares that were placed in boxes for the guests to take home as a wedding memento. Groom's cakes gradually died out and are no longer part of British weddings. However, the tradition has undergone a revival in the United States, where for many years the groom's cake has served as a wedding gift for guests. Modern groom's cakes

are often formed and decorated to depict the groom's hobby, for example a golf bag, a camera, a chess board.

Groom's cakes were never covered with icing, but Bride Cake covered with white icing first appeared sometime in the 17th century. After the cake was baked, it was covered with a pure white, smooth icing made with double refined sugar, egg whites, and orange-flower water. The mixture was beaten for two hours, then spread over the cake and dried in the oven until hard. A pure white color was much sought after for wedding cake icing because white icing meant that only the finest refined sugar had been used. Thus a pure white cake was a status symbol, as it displayed the family's wealth.

The late 1800s in Britain saw the introduction of a new tradition, with the first multi-tiered wedding cakes. These were impressive cakes: they were heavy because they were made with so much dried fruit, and highly decorated with icing and embellished with sugar flowers, doves, horseshoes and bells. The first multi-tiered cakes consisted of iced cakes stacked on top of each other after each cake had had time to harden. It was not until the beginning of the 20th century that the cake tiers were separated and supported by columns.

Twenty-first-century weddings are big business for Britain's wedding industry. Over 300,000 people get married each year and a wedding can cost thousands of pounds. The cost of the all-important wedding cake can be hundreds of pounds, depending on the dimension and design. It will be interesting to see whether wedding cakes continue to be popular at weddings.

QUESTION 1 – 6

Do the following statements agree with the information given in Reading Passage 1?

In boxes **1–6** on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN there is no information on this @Teaxona

1. Breaking a cake over the bride's head was the last part of an ancient Roman wedding ceremony.
2. Confetti is still made of nuts, dried fruit and honeyed almonds.
3. The groom's family used to supply the confetti.
4. It was considered lucky for a newly married couple to be able to kiss over a tower of spiced buns.
5. Only brides were allowed to eat Bride Pie.
6. The wedding cakes eaten in Britain today developed from Bride Cake of the 17th century.

Questions 7 – 9

Complete the table below. @Teaxona

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes **7–9** on your answer sheet.

Wedding cakes

17th century – Britain Bride Cake:

- expensive ingredients were a sign of wealth
- less expensive round cakes were made of **7**..... with currants in between and sugar on top they were baked on a hearth stone because not all homes had **8**.....

Now – United States – Groom's cake:

- guests receive pieces of the groom's cake
- cakes may represent the **9**..... of the groom

QUESTIONS 10 – 13

Label the diagram below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes **10–13** on your answer sheet.

- 10. A pure white cake represented the of the family.
- 11. Due to the amount of dried fruit, the cakes in 1800s were
- 12. In the early 1900s, were used to keep individual cakes apart.
- 13. The size of these cakes as well as their affects pricing.

Test 11: The early history of Scott and Bowne's cod liver oil

Cod liver oil is a type of fish oil used today for general health purposes, but northern European fishing communities used cod liver for centuries before the doctors and chemists of 19th-century Europe began to take an interest. Its manufacture was simple: the livers of the codfish were left for some days, then oil was taken from them. The oil grew darker according to how long the livers were left, resulting in three grades of oil: pale, light brown and dark brown.

Ludovicus Josephus de Jongh of the Netherlands produced the first extensive chemical analysis of cod liver oil in 1843. His studies of the three grades of oil led him to conclude that the light-brown oil was the most healthy. He attributed this superiority to the larger quantities of iodine, phosphate of chalk and volatile acids found in it.

In 1846, de Jongh traveled to Norway to obtain the purest oil available. By the 1850s, 'Dr. de Jong's Light Brown Cod Liver Oil' was marketed throughout Europe and exported to the United States. Each bottle had de Jongh's signature and stamped seal on it – a blue codfish on a red shield – guaranteeing that the product was 'put to the test of chemical analysis'. Advertising emphasized de Jongh's credentials as a doctor and chemist, and included testimonials from other men of science and medicine.

However, even the most enthusiastic supporters of cod liver oil admitted that the highly disagreeable taste and smell presented a significant obstacle to its use. De Jongh believed the problem of the oil's unpleasant taste and smell could be overcome with a little perseverance or, failing that, by following it with some fruit or biscuit, or glass of wine. But his recommendations appear not to have worked well. It was often combined with coffee, although a few people recommended taking the oil with tomato ketchup.

In 1873, Alfred B. Scott came to New York and, along with partner Samuel W. Bowne, began experimenting to produce a more pleasant preparation of cod liver. Three years later they established the firm of Scott and Bowne, and began marketing their product as Scott's Emulsion. Though not a doctor or pharmacist by training, Scott had the eye for opportunity that was necessary for achievement in business. Advertising, the two men believed, would propel their product to success. And so it did: by the 1890s Scott and Bowne had factories in five European countries, and were selling their emulsion throughout the Americas, Europe and Asia.

Scott got his oil for Scott's Emulsion directly from the Lofoten Islands in Norway, the world center of cod fishery – located above the Arctic Circle. The codfish streamed to the islands in early January to lay their eggs, and by the end of April were gone. The

Gulf Stream, the Arctic waters, and the Norwegian Fjords combined to create a perfect breeding group for the codfish and an unequaled fishing industry for the fishermen.

Scott and Bowne's first trademark, registered in 1879, included the initials P.P.P. and three words – 'Perfect, Permanent, Palatable'. The mark reflected that Scott's Emulsion was a perfect formula, a permanent emulsion (that is, one in which the ingredients would not separate), and most importantly, that it had a pleasant taste. 'Palatable as milk' became a key phrase in Scott's advertising.

A man with a fish on his back first appeared on Scott's Emulsion around 1884 and became Scott and Bowne's trademark in 1890. As Scott told it, he saw this fisherman with his record-breaking catch while on business in Norway. A photographer was quickly found to record the scene. Later, the photo was faithfully reproduced as a drawing, and registered as the company's trademark. In the drawing, the man stoops forward, glances out from under the brim of his hat, legs tensed under the weight of his load. A thick rope, wrapped round his waist, shoulders and hands, secures the load on his back – a huge fish with gaping mouth and glassy yellow eye, its tail sweeping the floor. The common codfish is recognizable by the brown and amber spots all over its body, the light stripe down its side, and the three dorsal fins. The words 'SCOTT'S EMULSION' appear in the title of the picture.

Trade cards and booklets featured the fisherman and his catch along with the words 'Scene taken from life on the coast of Norway' and 'The Codfish, weighing 156 pounds, was caught off the coast of Norway'. The realistic image, a direct reference to the natural source of the medicine, served as a reassurance of quality in a market that contained some impure, unsafe products.

Questions 1 – 8

Do the following statements agree with the information given in Reading Passage 1?
In boxes 1 – 8 on your answer sheet, write

TRUE	<i>if the statement agrees with the information</i>
FALSE	<i>if the statement contradicts the information</i>
NOT GIVEN	<i>if there is no information on this @Teaxona</i>

- 1 In the manufacturing process, cod livers left for the longest time produced the lightest type of oil.
- 2 A Dutch scientist called de Jongh suggested why one grade of cod liver oil was particularly healthy.
- 3 De Jongh was both a researcher in, and a supplier of, cod liver oil.
- 4 Many scientists tried to find a solution to the bad smell of cod liver oil.
- 5 The experimental methods of Scott and Bowne were much better than de Jongh's.
- 6 Scott was a trained chemist as well as a businessman.
- 7 Scott found a new location for cod fishing.
- 8 Cod around the Lofoten Islands could be caught during the first four months of the year.

Questions 9 – 13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 9 – 13 on your answer sheet.

Two versions of cod liver oil

De Jongh's version

to deal with the taste of the oil

- de Jongh suggested drinking some **9** after taking it
- others frequently added it to **10**

Scott and Bowne's version

trademarks and advertising:

- one slogan compared the product with **11**

the image of the man with a fish:

- contains
 - a man with a **12** around his body
 - a fish whose **13** is touching the ground

Test 12: The History of the Globe Theatre

The original Globe Theatre opened in autumn 1599 on the south bank of the River Thames, in London. It was one of several major theatres in the area, the others being the Swan, the Rose and the Hope. The Globe was one of the main playhouses of the Lord Chamberlain's Men, later called the King's Men, a theatrical company to which the playwright William Shakespeare belonged. Most of Shakespeare's post-1599 plays were staged at the Globe, including *Julius Caesar*, *Macbeth*, *Othello*, *King Lear* and *Hamlet*.

The Globe was built in 1576, using timber from an earlier theatre in Shoreditch, east London, by James Burbage, the father of one of the actors. Called simply The Theatre, this structure was taken down after a 20-year lease on its land expired, and the timber from it was then transported over the Thames to construct the Globe.

The Globe Theatre was highly successful. Since there was little artificial light, performances were held in the afternoon, usually beginning in May. Flags waving in the wind advertised plays by the King's Men, and all about the grounds of the theatre, playgoers would be buying merchandise and refreshments from stalls. People from all walks of life could enjoy the tragedies and comedies of Shakespeare. Both men and women would attend the plays, but the well-off would frequently wear masks to conceal their identity.

After years of success, the Globe was destroyed by fire on 29 June 1613, during a performance of *Henry VIII*. A theatrical cannon, set off during the performance, misfired, igniting the building's roof and wooden beams. According to one of the few surviving contemporary accounts of the event, no one was hurt except a man who put out his burning trousers with a bottle of ale. A new Globe was created by the theatre company on the foundations of its predecessor before Shakespeare's death in 1616.

However, after the Puritans gained control of Parliament, their campaign to enforce a higher standard of morality meant the end of the Globe. Their strict views gave rise to restrictions on many social activities, and like all the other theatres in London, the Globe was closed down in 1642. In 1644, the Puritans destroyed the Globe altogether to make room for housing. Its exact location was forgotten and remained unknown until remnants of its foundations were discovered in 1989 near a row of eighteenth-century buildings. There may be further remains under these buildings but they are themselves listed as historical buildings, and currently may not be disturbed by archaeologists.

The Globe's shape and size have been the subject of scholarly inquiry over the last two centuries. The evidence suggests that it was a three-storey, 100-foot-wide, open-air amphitheatre. It was probably a polygon in shape, with twenty sides. At the base of the stage there was an area called the "pit", where spectators (called "groundlings") would either stand or sit on the ground to watch the performance. They paid one penny to see

the play, and the box into which they put their entrance fee gave rise to the term “box office”.

A rectangular platform, also known as an “apron stage”, thrust out into the middle of the open-air yard. On this, there was a trap door for use by performers to enter from the “cellarage” area beneath the stage. Large columns on either side of the stage supported a roof over the rear portion of the stage. This would have been constructed using thatch. The inner ceiling was called the “heavens”, and was possibly painted with images of the sky. An opening in the heavens enabled performers to descend, using some form of rope and harness. The back wall of the stage had three doors on the first floor and a balcony on the second. The doors led into the “tiring house” where the actors dressed and awaited their cue. The balcony was where the musicians performed and could also be used for scenes requiring an upper space, such as the balcony scene in Shakespeare’s *Romeo and Juliet*. In addition, it could be used as the “Lord’s Room”, where higher-paying audience members could be seated.

When the foundations of Shakespeare’s Globe were discovered in London’s Bankside district, public interest in building a modern version was reignited. Workers began construction in 1993, close to the site of the original. Completed in 1996, the current Globe Theatre was officially opened on 12 June 1997 with a production of *Henry V*. The reconstruction was carefully researched, so that the new building would be as faithful a replica as possible. It had the first thatched roof permitted in the city since the Great Fire of London in 1666. Modernisations include the addition of sprinklers on the roof to protect against fire. Due to modern health and safety regulations, only 1,300 people can attend a show, under half the estimated 3,000 of Shakespeare’s time.

Tickets to stand in the open — no sitting allowed in this section — are available for every performance at £5 each. The only covered parts of the amphitheatre are the stage and the more expensive seated areas. Plays are normally performed between May and October, and in winter the theatre is used for educational purposes.

Questions 1–6 @Teaxona

Do the following statements agree with the information in Reading Passage 1?

In boxes **1–6** on your answer sheet, write:

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this @Teaxona

1. The Lord Chamberlain's Men performed plays only in the Globe Theatre.
2. Recycled wood was used in the construction of the original Globe.
3. Performances at the Globe were so popular that advance booking was necessary.
4. Some people who watched the plays tried to avoid recognition.
5. When the original Globe burned down, it was rebuilt on the same site.
6. The Globe was the first theatre to be closed down by the Puritans.

Questions 8–13

Complete the sentences below.

Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer. @Teaxona

Write your answers in boxes **8–13** on your answer sheet.

The Globe

Parts of the original Globe

- The **7** _____ — for members of the audience (one penny per person)
- The apron stage — a protruding platform
- The trap door — giving access to the stage from below
- Roof made of **8** _____
- The **9** _____ — for performers to change into costumes
- Balcony used by **10** _____ or wealthy spectators

The modern Globe

- 1993–1996 — the modern Globe was built

The building

- It was designed to be a mainly accurate replica
- **11** _____ were a new safety feature
- Audience capacity was reduced by more than **12** _____

The performances

- The price of outdoor tickets is fixed
- Tickets for seats in the **13** _____ spaces cost more
- There are no performances during winter

Test 13: The History of Dolls

What is today a simple children's toy has a surprisingly rich history. Dolls have been a part of humankind for thousands of years. Often depicting religious figures, or used as playthings, early dolls were probably made from primitive materials such as clay, fur, or wood. Dolls constructed of flat pieces of wood, painted with various designs, and with hair made of clay, have often been found in Egyptian graves dating back to 2000 BC. Egyptian tombs of wealthy families have included pottery dolls. Dolls being placed in these graves leads some to believe that they were cherished possessions.

Girls from ancient Greece and Rome offered their wooden dolls to goddesses after they were too 'grown-up' to play with dolls. Most ancient dolls that were found in tombs were very simple creations, often made from such materials as clay, rags, wood, or bone. Some of the more unique dolls were made with ivory or wax. The main goal was to make the doll as lifelike as possible. That ideal led to the creation of dolls with movable limbs and removable garments, dating back to 600 BC.

Following the era of the ancient dolls, Europe became a major hub for doll production. These dolls were primarily made of wood. Fewer than 30 examples of primitive wooden stump dolls from England survive today. The Grodnertal area of Germany produced many peg wooden dolls, a type of doll that has very simple peg joints and resembles a clothespin (a device for hanging washing on a clothesline). An alternative to wood was developed in the 1800s.

'Composition' is a collective term for mixtures of pulped wood or paper that were used to make doll heads and bodies. These mixtures were moulded under pressure, creating a durable doll that could be mass produced. Manufacturers closely guarded the recipes for their mixtures, sometimes using strange ingredients like ash or eggshells. Papier-mâché, a type of composition, was one of the most popular mixtures.

In addition to wooden dolls, wax dolls grew in popularity in the 17th and 18th centuries. Much in Germany was a major manufacturing center for wax dolls. Wax dollmakers would model a doll's head in wax or clay, and then cover it with plaster to create a mould. Then they would pour melted wax into the cast. The wax for the head would be very thin, no more than 3 mm. Some of the most distinctive wax dolls were created in England between 1850 and 1930. One of the first dolls that portrayed a baby was made in England from wax at the basinsion of the 10th century.

Around the same time porcelain became popular. It is made by firing special clays in a kiln at more than 2372 degrees Fahrenheit (1300°C), and only a few clays can withstand firing at such high temperatures. Porcelain is used generically to refer to both china and bisque dolls; china is glazed, whereas bisque is unglazed. Germany, France, and Denmark started creating china heads for dolls in the 1840s. These china heads were

replaced in the 1860s by ones made of bisque. Bisque, which is porcelain fired twice with colour added to it after the first firing, looked more like skin than china did.

In France, the bébé was popular in the 1880s, and it has become a highly sought after doll today. The bébé, first made in the 1850s, was different from its predecessors because it depicted a younger girl. Until then, most French dolls were representations of adults. Although the French dolls were unrivalled in their artistry, German bisque dolls became quite popular because they were not as expensive. Kammer & Reinhardt introduced a bisque character doll in the 1900s, starting a trend of creating realistic dolls.

For many centuries, rag dolls were made by mothers for their children. The term 'rag doll' refers generically to dolls made of any fabric. 'Cloth doll' refers to a subset of rag dolls made of linen or cotton. Commercially produced rag dolls were first introduced in the 1850s by English and American manufacturers. Although not as sophisticated as dolls made from other materials, rag dolls were well loved, often as a child's first toy.

Dollmaking did not become an industry in the United States until after the Civil War in the 1860s. Doll production was concentrated in the New England region of the United States, with dolls made from a variety of materials such as leather, rubber, papier-mâché, and cloth. Celluloid was developed in the state of New Jersey in the late 1860s and was used to manufacture dolls until the mid-1950s. German, French, American, and Japanese factories churned out cheaply produced celluloid dolls in mass quantities. However, celluloid fell out of favour because of its extreme flammability and propensity to fade in bright light.

After World War I, doll makers experimented with plastics. Hard plastic dolls were manufactured in the 1940s. They resembled composition dolls, but they were much more durable. Other materials used in doll manufacturing included rubber, foam rubber, and vinyl in the 1950s and 1960s. Vinyl changed doll making, allowing doll makers to root hair into the head, rather than using wigs or painting the hair. Although most dolls are now mass-manufactured using these modern materials, many modern doll makers are still using the techniques and craftsmanship of earlier centuries, creating limited-edition dolls that are highly prized by collectors. These handmade dolls often reflect traditional methods, emphasising fine detail and artistic quality over mass production.

Questions 1–6

Complete the notes below.

Choose **ONE WORD ONLY** for each answer.

Earliest known dolls

- Represented religious figures
- Used as toys
- Egypt 2000 BC
- Bodies were made of **1**
- **2** was used for the hair

Ancient Greece and Rome

- Dolls were given to **3** by older girls

600 BC

- realistic dolls had separate clothes and **4** that could be put in different positions

17th and 18th centuries

- Dolls made of **5**
- Moulds made of plaster

1800s

- **6** became more common
- New manufacturing process developed
- Recipes for these mixtures kept secret

Questions 7–13

Do the following statements agree with the information in the Reading Passage?

Write:

TRUE – if the statement agrees with the information

FALSE – if it contradicts the information

NOT GIVEN – if there is no information @Teaxona

7. Bisque dolls appear less realistic than dolls made of china.
8. French dolls tended to cost more than German bisque dolls.
9. The first rag dolls were manufactured in the 1850s.
10. Only dolls made of cotton or linen are classified as cloth dolls.
11. Dolls made of celluloid tended to lose their colour.
12. Composition dolls lasted longer than the plastic dolls that were made in the 1940s.
13. Doll collectors prefer a doll to be dressed in its original clothing.

Test 14: The History of Building Telegraph Lines

A

The idea of electrical communication seems to have begun as long ago as 1746 when about 200 monks at a monastery in Paris arranged themselves in a line over a mile long, each holding ends of 25 ft iron wires. The abbot, also a scientist, discharged a primitive electrical battery into the wire, giving all the monks a simultaneous electrical shock. “This all sounds very silly, but is in fact extremely important because, firstly, they all said ‘ow’ which showed that you were sending a signal right along the line; and, secondly, they all said ‘ow’ at the same time, and that meant that you were sending the signal very quickly,” explains Tom Standage, author of *The Victorian Internet* and technology editor at the *Economist*. Given a more humane detection system, this could be a way of signaling over long distances.

B

With wars in Europe and colonies beyond, such a signaling system was urgently needed. All sorts of electrical possibilities were proposed, some of them quite ridiculous. Two Englishmen, William Cooke and Charles Wheatstone came up with a system in which dials were made to point at different letters, but that involved five wires and would have been expensive to construct.

C

Much simpler was that of an American, Samuel Morse, whose system only required a single wire to send a code of dots and dashes. At first, it was imagined that only a few highly skilled encoders would be able to use it but it soon became clear that many people could become proficient in Morse code. A system of lines strung on telegraph poles began to spread in Europe and America.

D

The next problem was to cross the sea. Britain, as an island with an empire, led the way. Any such cable had to be insulated and the first breakthrough came with the discovery that a rubber-like latex from a tropical tree on the Malay Peninsula could do the trick. It was called gutta-percha. The first attempt at a cross-channel cable came in 1850. With thin wire and thick insulation, it floated and had to be weighed down with a lead pipe.

E

It never worked well as the effect of water on its electrical properties was not understood, and it is reputed that a French fisherman hooked out a section and took it home as a strange new form of seaweed. The cable was too big for a single boat so two had to start in the middle of the Atlantic, join their cables and sail in opposite directions. Amazingly, they succeeded in 1858, and this enabled Queen Victoria to send a telegraph message to President Buchanan. However, the 98-word message

took more than 19 hours to send and a misguided attempt to increase the speed by increasing the voltage resulted in the failure of the line a week later.

F

By 1870, a submarine cable was heading towards Australia. It seemed likely that it would come ashore at the northern port of Darwin from where it might connect around the coast to Queensland and New South Wales. It was an undertaking more ambitious than spanning an ocean. Flocks of sheep had to be driven with the 400 workers to provide food. They needed horses and bullock carts and, for the parched interior, camels. In the north, tropical rains left the teams flooded. In the centre, it seemed that they would die of thirst. One critical section in the red heart of Australia involved finding a route through the McDonnell mountain range and then finding water on the other side.

G

The water was not only essential for the construction team. There had to be telegraph repeater stations every few hundred miles to boost the signal and the staff obviously had to have a supply of water. Just as one mapping team was about to give up and resort to drinking brackish water, some aboriginals took pity on them. Altogether, 40,000 telegraph poles were used in the Australian overland wire. Some were cut from trees. Where there were no trees, or where termites ate the wood, steel poles were imported.

H

On Thursday, August 22, 1872, the overland line was completed and the first messages could be sent across the continent; and within a few months, Australia was at last in direct contact with England via the submarine cable, too. The line remained in service to bring news of the Japanese attack on Darwin in 1942. It could cost several pounds to send a message and it might take several hours for it to reach its destination on the other side of the globe, but the world would never be the same again. Governments could be in touch with their colonies. Traders could send cargoes based on demand and the latest prices. Newspapers could publish news that had just happened and was not many months old.

Questions 1–6

Do the following statements agree with the information in the passage? Write:

TRUE — if the statement agrees with the passage

FALSE — if the passage contradicts the statement

NOT GIVEN — if there is no information on this @Teaxona

1. In the research of French scientists, the metal lines were used to send a message.
2. Monks gave the abbots an electrical shock at the same time, which constitutes the exploration of the long-distance signalling.
3. Using Morse Code to send message needs to simplify the message firstly.
4. Morse was a famous inventor before he invented the code.
5. The water is significant to early telegraph repeater on the continent.
6. US Government offered fund to the first overland line across the continent.

Questions 7–14

Complete the summary below. Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

The Development of Telegraph Cable Systems

The earliest telegraph system developed by Charles Wheatstone was unreliable at first because it was **7.** _____. When engineers later attempted to lay cable across the sea, they used **8.** _____ to provide insulation. To ensure that the cable would sink to the ocean floor, British pioneers added **9.** _____ to increase its weight. However, part of the early cable in shallow waters was damaged because a fisherman accidentally thought it was **10.** _____.

When the first transatlantic line was completed, **11.** _____ received the very first message from the Queen. In other regions, engineers faced additional challenges. In desert areas, the cable was transported using **12.** _____. In the red heart of Australia, workers needed to locate a **13.** _____ among the mountains.

Test 15: The history (impact) of the Potato

Jeff Chapman relates the story of history's most important vegetable

A The potato was first **cultivated** in South America between three and seven thousand years ago, though scientists believe they may have grown wild in the region as long as 13,000 years ago. The genetic patterns of potato distribution indicate that the potato probably originated in the mountainous west-central region of the continent. Early Spanish chroniclers who misused the Indian word *batata* (sweet potato) as the name for the potato noted the importance of the tuber to the Incan Empire. The Incas had learned to preserve the potato for storage by dehydrating and mashing potatoes into a substance called *Chuño*. It could be stored in a room for up to 10 years, providing excellent insurance against possible crop failures. As well as using the food as a staple crop, the Incas thought potatoes made childbirth easier and used it to treat injuries.

B The Spanish conquistadors first encountered the potato when they arrived in Peru in 1532 in search of gold, and noted Inca miners eating *Chuño*. At the time the Spaniards failed to realise that the potato represented a far more important treasure than either silver or gold, but they did gradually begin to use potatoes as basic rations aboard their ships. After the arrival of the potato in Spain in 1570, a few Spanish farmers began to cultivate them on a small scale, mostly as food for livestock. Throughout Europe, potatoes were regarded with suspicion, **distaste** and fear. Generally considered to be **unfit** for human consumption, they were used only as animal fodder and sustenance for the starving. In northern Europe, potatoes were primarily grown in botanical gardens as an exotic novelty. Even **peasants** refused to eat from a plant that produced ugly, misshapen **tubers** and that had come from a **heathen** civilisation. Some felt that the potato plant's **resemblance** to plants in the **nightshade** family **hinted** that it was the creation of **witches** or devils.

C In meat-loving England, farmers and urban workers regarded potatoes with extreme distaste. In 1662, the Royal Society recommended the cultivation of the tuber to the English government and the nation, but this recommendation had little impact. Potatoes did not become a staple until, during the food shortages associated with the Revolutionary Wars, the English government began to officially encourage potato cultivation. In 1795, the Board of Agriculture issued a pamphlet entitled *Hints Respecting the Culture and Use of Potatoes*; this was followed shortly by pro-potato editorials and potato recipes in *The Times*. Gradually, the lower classes began to follow the lead of the upper classes.

D A similar pattern emerged across the English Channel in the Netherlands, Belgium and France. While the potato slowly gained ground in eastern France (where it was often the only crop remaining after marauding soldiers plundered wheat fields and vineyards), it did not achieve widespread acceptance until the late 1700s. The peasants remained suspicious, in spite of a 1771 paper from the Faculté de Paris testifying that the potato was not harmful but beneficial. The people began to overcome their distaste when the plant received the royal seal of approval: Louis XVI began to sport a potato flower in his buttonhole, and Marie-Antoinette wore the purple potato blossom in her hair. Frederick the Great of Prussia saw the potato's

potential to help feed his nation and lower the price of bread, but faced the challenge of overcoming the people's prejudice against the plant.

E When he issued a 1774 order for his subjects to grow potatoes as protection against famine, the town of Kolberg replied: “The things have neither smell nor taste, not even the dogs will eat them, so what use are they to us?” Trying a less direct approach to encourage his subjects to begin planting potatoes, Frederick used a bit of reverse psychology: he planted a royal field of potato plants and stationed a heavy guard to protect this field from thieves. Nearby peasants naturally assumed that anything worth guarding was worth stealing, and so snuck into the field and snatched the plants for their home gardens. Of course, this was entirely in line with Frederick's wishes.

F Historians debate whether the potato was primarily a cause or an effect of the huge population boom in industrial-era England and Wales. Prior to 1800, the English diet had consisted primarily of meat, supplemented by bread, butter and cheese. Few vegetables were consumed, most vegetables being regarded as nutritionally worthless and potentially harmful. This view began to change gradually in the late 1700s. The Industrial Revolution was drawing an ever-increasing percentage of the populace into crowded cities, where only the richest could afford homes with ovens or coal storage rooms, and people were working 12–16 hour days which left them with little time or energy to prepare food. High-yielding, easily prepared potato crops were the obvious solution to England's food problems.

G Whereas most of their neighbours regarded the potato with suspicion and had to be persuaded to use it by the upper classes, the Irish peasantry embraced the tuber more passionately than anyone since the Incas. The potato was well suited to the Irish soil and climate, and its high yield suited the most important concern of most Irish farmers: to feed their families. The most dramatic example of the potato's potential to alter population patterns occurred in Ireland, where the potato had become a staple by 1800.

H The Irish population doubled to eight million between 1780 and 1841, this without any significant expansion of industry or reform of agricultural techniques beyond the widespread cultivation of the potato. Though Irish land-holding practices were primitive in comparison with those of England, the potato's high yields allowed even the poorest farmers to produce more healthy food than they needed with scarcely any investment or hard labour. Even children could easily plant, harvest and cook potatoes, which of course required no threshing, curing or grinding. The abundance provided by potatoes greatly decreased infant mortality and encouraged early marriage.

Questions 1-5

Do the following statements agree with the information given in Reading Passage 1? In boxes 1-5 on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this @Teaxona

1. The early Spanish called potato as the Incan name 'Chuño'.
2. The purpose of Spanish coming to Peru was to find potatoes.
3. The Spanish believed that the potato has the same nutrients as other vegetables.
4. Peasants at that time did not like to eat potatoes because they were ugly.
5. The popularity of potatoes in the UK was due to food shortages during the war.

Questions 6-13

Complete the sentences below.

Choose **ONE WORD ONLY** from the passage for each answer. Write your answers in boxes 6-13 on your answer sheet.

6	In France, people started to overcome their disgusting about potatoes because the King put a potato in his button hole.
7	Frederick realised the potential of potato but he had to handle the against potatoes from ordinary people.
8	The King of Prussia adopted some psychology to make people accept potatoes.
9	Before 1800, the English people preferred eating with bread, butter and cheese.
10	The obvious way to deal with England food problems was to grow high yielding potato
11	The Irish and climate suited potatoes well.
12	Between 1780 and 1841, based on the of the potatoes, the Irish population doubled to eight million.
13	The potato's high yields helped the poorest farmers to produce more healthy food almost without or hard physical work.

Test 16: The History of American dance

The birth of modern American dance occurred in the first years of the twentieth century. And, perhaps unusually for academics, dance historians hold remarkably similar views when it comes to identifying early 1900s, we can see that dancers quite deliberately moved away from previous approaches. This included rejecting both the formal moves of ballet dancing and the entertainment of vaudeville dancing. As a result, dancers began the new century with a fresh start. One important figure at this time was Loie Fuller, who performed largely with her arms, perhaps because she had limited dance training. Fuller emphasized visual effects rather than storytelling, and pioneered the use of artificial lighting to create shadows while dancing.

Perhaps most influential in the early years was Isadora Duncan, who was well known in both America and Europe. Duncan refused to wear elaborate costumes, preferring to dance in plain dresses and bare feet. She is also notable for preferring music written by classical composers such as Chopin and Beethoven, rather than contemporary compositions. At a similar time, Ruth St Denis was bringing the dance training academy with her husband with the intention of passing on her approach and style to the next generation of American dancers.

By the 1920s, the modern dance movement in America was well established. Audiences were enthusiastic and dancers were increasingly prepared to experiment with new ideas. Martha Graham was one of an important group who emerged in New York. Graham looked within herself to find her dance style, examining how her body moved as she breathed, but also observing the patterns made by her limbs when walking in order to find a new, naturalistic approach to dance. Doris Humphrey wanted her dance to reflect her personal experience of American life. She explored the concept of gravity, allowing her body to fall, only to recover at the last moment. Her book *The Art of Making Dances*, which detailed her approach to dance composition, was highly influential with later generations of dancers.

By the 1930s, modern dance was becoming an accepted, respectable art form. Universities such as Bennington College included modern dance in their performing arts programmes for the first time. In the 1940s, German-born dancer Hanya Holm embraced the changing times by including modern dance in mainstream musicals on the Broadway stage. Among Holm's many other innovations was bringing her own humour to these performances – audiences adored it.

Modern American dance has seldom stood still. Each new generation of dancers either developed the techniques of their teachers or rejected them outright. So by the 1950s the techniques of traditional European ballet dancing were again influential. This was certainly true of Erick Hawkins, who also incorporated Native American and Asian

styles. Similarly, Merce Cunningham emphasized the leg actions and flexibility of the spine associated with ballet moves. Paul Taylor preferred his dance to reflect the experiences and interactions of ordinary people going about their everyday lives. Taylor's career was the subject of a documentary that provided valuable insights into this period of dance.

The middle decades of the 20th century were certainly a dynamic time. Increasingly, the modern dance movement recognized and reflected the fact that America was a multi-racial, multi-cultural society. Katherine Dunham, an anthropology graduate, used movements from Pacific, African and Caribbean dance to create her unique style. Pearl Primus was another champion of African dance, which she passed on through her dance school in New York. After retirement she travelled widely to universities throughout America lecturing on ethnic dance, which became her main priority.

Modern dance since the 1980s has become a mix of multiple forms of dance, as well as art more generally. For example, Mark Morris's hugely popular work *The Hard Nut* includes sensational costumes and a stage design inspired by the comics he'd always enjoyed. Another innovator has been Ohad Naharin, who studied in New York and has worked internationally. Naharin's 'Gaga' style is characterized by highly flexible limbs and backbones, while in rehearsal his dancers have no mirrors, feeling their movements from within themselves, a break from traditional dance custom. In many ways it was a fitting end to a 100-year period that had witnessed a transformation in dance. The emergence of modern American dance was very much a 20th –century phenomenon. The style drew on influences from home and abroad and in turn went on to influence global dance culture.

Questions 1–6

Do the following statements agree with the information given in Reading Passage 1?

In boxes 1–6 on your answer sheet, write

TRUE	<i>if the statement agrees with the information</i>
FALSE	<i>if the statement contradicts the information</i>
NOT GIVEN	<i>if there is no information on this @Teaxona</i>

- 1 Dance historians agree about the development of modern American dance.
- 2 Dancers in the early 1900s tended to copy the styles of earlier dancers.
- 3 Loie Fuller preferred to dance alone on stage.
- 4 Isadora Duncan wore complicated clothing when dancing.
- 5 Some dancers criticized Isadora Duncan for her choice of music.
- 6 Ruth St Denis wished to educate others in her style of dancing.

Questions 7–10

Complete the notes below.

Choose **ONE WORD AND/OR A NUMBER** from the passage for each answer.

Write your answers in boxes 7–10 on your answer sheet.

Developments in Modern American Dance

1920s-1940s

- Martha Graham based her dance on human actions such as breathing and **7**
- Doris Humphrey wrote an important **8** about her ideas.
- Dance became a respectable subject to study at university.
- Hanya Holm introduced **9** into dance and musicals.

1950s-1970s

- Eric Hawkins and Merce Cunningham reintroduced some ballet techniques.
- An influential **10** outlined the working life of Paul Taylor.

Questions 11–13

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

11. After stopping her dance career, Pearl Primus concentrated on

12. A major inspiration for Mark Morris's The Hard Nut was

13. Ohad Naharin's dancers train without using

Test 17: The History of Salt

A

Salt is so simple and plentiful that we almost take it for granted. In chemical terms, salt is the combination of a sodium ion with a chloride ion, making it one of the most basic molecules on earth. It is also one of the most plentiful: it has been estimated that salt deposits under the state of Kansas alone could supply the entire world's needs for the next 250,000 years.

B

But salt is also an essential element. Without it, life itself would be impossible since the human body requires the mineral in order to function properly. The concentration of sodium ions in the blood is directly related to the regulation of safe body fluid levels. And while we are all familiar with its many uses in cooking, we may not be aware that this element is used in some 14,000 commercial applications. From manufacturing pulp and paper to setting dyes in textiles and fabric, from producing soaps and detergents to making our roads safe in winter, salt plays an essential part in our daily lives.

C

Salt has a long and influential role in world history. From the dawn of civilization, it has been a key factor in economic, religious, social and political development. In every corner of the world, it has been the subject of superstition, folklore, and warfare, and has even been used as currency.

D

As a precious and portable commodity, salt has long been a cornerstone of economies throughout history. In fact, researcher M.R. Bloch conjectured that civilization began along the edges of the desert because of the natural surface deposits of salt found there. Bloch also believed that the first war – likely fought near the ancient city of Assault on the Jordan River – could have been fought over the city's precious supplies of the mineral.

E

In 2200 BC, the Chinese emperor Hsia Yu levied one of the first known taxes. He taxed salt. In Tibet, Marco Polo noted that tiny cakes of salt were pressed with images of the Grand Khan to be used as coins, and to this day among the nomads of Ethiopia's Danakil Plains, it is still used as money. Greek slave traders often bartered it for slaves, giving rise to the expression that someone was "not worth his salt." Roman legionnaires were paid in salt – a *salarium*, the Latin origin of the word "salary."

F

Merchants in 12th-century Timbuktu – the gateway to the Sahara Desert and the seat of scholars – valued this mineral as highly as books and gold. In France, Charles of Anjou

levied the *gabelle*, a salt tax, in 1259 to finance his conquest of the Kingdom of Naples. Outrage over the *gabelle* fueled the French Revolution. Though the revolutionaries eliminated the tax shortly after Louis XVI, the Republic of France re-established the *gabelle* in the early 19th century; only in 1946 was it removed from the books.

G

The Erie Canal, an engineering marvel that connected the Great Lakes to New York's Hudson River in 1825, was called "the ditch that salt built." Salt tax revenues paid for almost half of the cost of construction of the canal. The British monarchy supported itself with high salt taxes, leading to a bustling black market for the white crystal. In 1785, the Earl of Dundonald wrote that every year in England, 10,000 people were arrested for salt smuggling. And protesting against British rule in 1930, Mahatma Gandhi led a 200-mile march to the Arabian Ocean to collect untaxed salt for India's poor.

H

In religion and culture, salt long held an important place with Greek worshippers consecrating it in their rituals. Further, in Buddhist tradition, salt repels evil spirits, which is why it is customary to throw it over your shoulder before entering your house after a funeral: it scares off any evil spirits that may be clinging to your back. Shinto religion also uses it to purify an area. Before sumo wrestlers enter the ring for a match – which is, in reality, an elaborate Shinto rite – a handful is thrown into the center to drive off malevolent spirits.

I

In the Southwest of the United States, the Pueblo worship the Salt Mother. Other native tribes had significant restrictions on who was permitted to eat salt. Hopi legend holds that the angry Warrior Twins punished mankind by placing valuable salt deposits far from civilization, requiring hard work and bravery to harvest the precious mineral. Today, a gift of salt endures in India as a potent symbol of good luck and a reference to Mahatma Gandhi's liberation of India.

J

The effects of salt deficiency are highlighted in times of war, when human bodies and national economies are strained to their limits. Thousands of Napoleon's troops died during the French retreat from Moscow due to inadequate wound healing and lowered resistance to disease – the results of salt deficiency.

Questions 1–3

Choose **THREE** letters, A–H.

Write your answers in boxes **1–3** on your answer sheet.

Which **THREE** statements are true of salt?

- A. A number of cities take their name from the word salt.
- B. Salt contributed to the French Revolution.
- C. The religious uses of salt are countless.
- D. Salt has been produced in China for less than 2000 years.
- E. There are many commercial applications for salt.
- F. Salt deposits in the state of Kansas are vast.
- G. Salt has few industrial uses nowadays.
- H. Slaves used salt as a currency.

Questions 4–8

Complete the summary below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Write your answers in boxes **4–8** on your answer sheet.

Salt is such an **4** _____ that people would not be able to live without it. As well as its uses in cooking, this basic mineral has thousands of business **5** _____, ranging from making paper to the manufacture of soap. Being a prized and **6** _____, it has played a major part in the economies of many countries. As such, salt has not only led to war, but has also been used to raise **7** _____ by governments in many parts of the world. There are also many instances of its place in religion and culture, being used as a means to get rid of evil **8** _____.

Questions 9–13

Do the following statements agree with the information in Reading Passage 1?

*In boxes **9–13** on your answer sheet, write:*

TRUE – *if the statement agrees with the information*

FALSE – *if the statement contradicts the information*

NOT GIVEN – *if there is no information on this @Teaxona*

- 9.** It has been suggested that salt was responsible for the first war.
- 10.** The first tax on salt was imposed by a Chinese emperor.
- 11.** Salt is no longer used as a form of currency.
- 12.** Most of the money for the construction of the Erie Canal came from salt taxes.
- 13.** A lack of salt is connected with the deaths of some soldiers.

Test 18: A Brief History of Rubber

A

Rubber is one of the most important products to come out of the rainforest. Though indigenous rainforest dwellers of South America have been using rubber for generations, it was not until 1839 that rubber had its first practical application in the industrial world. In that year, Charles Goodyear accidentally dropped rubber and sulfur on a hot stovetop, causing it to char like leather yet remain plastic and elastic. Vulcanization, a refined version of this process, transformed the white sap from the bark of the Heave tree into an essential product for the industrial age.

B

With the invention of the automobile in the late 19th century, the rubber boom began. As demand for rubber soared small dumpy river towns like Manaus, Brazil, were transformed into over night into bustling centers of commerce. Manaus, situated on the Amazon where it is met by Rio Negro, became the opulent heart of the rubber trade. Within a few short years Manaus had Brazil's first telephone system, 161 miles of streetcar tracks, and an electric grid for a city of a million, though it had a population of only 40,000.

C

The opulence of the rubber barons could only be exceeded by their brutality. Wild Heave trees, like all primary rainforest trees are widely dispersed, with an adaptation that protects species from the South American leaf blight which easily spreads through and decimates plantations. Thus, to make a profit, barons had to acquire control over huge tracts of land. Most did so by hiring their own private armies to defend their claims, acquire new land, and capture native laborers. As the Indians died, production soared.

D

The Brazilian rubber market was crushed by the rapid development of the more efficient rubber plantations of Southeast Asia. However, the prospects of developing plantations did not begin on a high note. Rubber seeds, rich with oil and latex, could not survive the long Atlantic journey from Brazil. Finally, in 1876, an English planter, Henry Wickham, collected 70,000 seeds and shipped them to England. 2800 of the seeds germinated and were sent to Colombo, Ceylon (present day Sri Lanka). After several false starts, including one planter in northern Borneo who felled his plantation after finding no rubber balls hanging from the branches, the prospects were grim. One major obstacle was the success of tea and coffee gave planters no reason to try an untested crop.

E

Finally in 1895, Henry Ridley, head of Singapore's botanical garden, persuaded two coffee growers to plant two acres of Heave tress. Twelve years later more than 300,000 ha of rubber grew in plantations in Ceylon and Malaya. New innovations increased efficiency and production doubled every two years. Rubber could be produced at only a fraction of the cost of collecting wild rubber in Brazil. By 1910, Brazilian production had fallen to 50%. In 1914, Brazil's market share was down around 30%; 1918 -20%, and 1940 -1.3%.

F

However, the Second World War threatened to shift the rubber wealth. With Japan occupying prime rubber producing areas in Southeast Asia, the US feared it would run out of the vital material. Every tire, hose, seal, valve, and inch of wiring required rubber. The rubber Development Corporation, the chief overseer of rubber acquisition, sought out other sources including establishing a rubber program that sent intrepid explorers into the Amazon seeking rubber specimen that would be used to produce high yields, superior product, and possibility of resistance against leaf blight. The ultimate goal of the program was to establish rubber plantations close to home. In addition to searching the Amazon and establishing experimental plantations in Latin America, the program came up with some novel plans to produce rubber. Extensive work on synthetic rubber yielded a product that, in time, economists predicted would replace natural rubber. By 1964 synthetic rubber made up 75% of the market.

G

However, the situation changed drastically with the OPEC oil embargo of 1973 which doubled the price of synthetic rubber and made oil consumers more conscious of their gas mileage. The concern over gas mileage brought unexpected threat to the synthetic market: the wide-spread adoption of the radial tire. The radial tire replaced the simple bias tires (which made up 90% of the market only 5 year earlier) and within a few years virtually all cars were rolling in radials. Synthetic rubber did not have the strength for radials; only natural rubber could provide the required sturdiness. By 1993 natural rubber had recaptured 39% of the domestic market. Today nearly 50% of every auto tire and 100% of all aircraft tires are made of natural rubber. 85% of this rubber is imported from Southeast Asia meaning that the US is highly susceptible to disruptions caused by an embargo or worse, the unintentional or intentional introduction of leaf blight into plantations. None of the trees in plantations across Southeast Asia has resistance to blight so a single act of biological terrorism, the systematic introduction of fungal spores so small as to be readily concealed in a shoe, could wipe out the plantations, shutting down production of natural rubber for at least a decade. It is difficult to think of any other raw material that is as vital and vulnerable.

Questions 1-7

Reading Passage 1 contains 7 paragraphs **A-G**. @Teaxona

Which paragraphs state the following information?

*Write the appropriate letters **A-G** in boxes **1-7** on your answer sheet.*

***N.B** you may use any letter more than once.*

- 1 The extensive acceptance of radial tires.
- 2 Searching for new specimens to overcome leaf blight
- 3 The first trading center for the rubber business.
- 4 Asia dominated the rubber market year by year
- 5 Rubber seeds are vulnerable to long distance transport
- 6 Individual wealth accumulated by rubber trading
- 7 Natural rubber gave way to its replacement

Questions 8-13

Complete the summary below based on paragraph G

*Use **NO MORE THAN TWO WORDS** from the passage for each answer.*

OPEC doubled revenue from synthetic rubber and made oil consuming nations more attentive of their **8** _____. This brought threats to the synthetic market by espousing the **9** _____, which would replace all the simple bias tires within a few years. Because **10** _____ is the only material that provides the entailed toughness, synthetic rubber lost significant market share. The US industry is very fragile to disruptions caused by an **11** _____. What's even worse, since the rubber trees in plantations across Southeast Asia cannot withstand **12** _____, the small fungal spores could be so dangerous as to shut down production of natural rubber for a decade. Rubber, hence, is the most important and **13** _____ raw material in the world.

Test 19: The Cacao: A Sweet History

A

Most people today think of chocolate as something sweet to eat or drink that can be easily found in stores around the world. It might surprise you that chocolate was once highly treasured. The tasty secret of the cacao (Kah Kow) tree was discovered 2,000 years ago in the tropical rainforests of the Americas. The story of how chocolate grew from a local Mesoamerican beverage into a global sweet encompasses many cultures and continents.

B

Historians believe the Maya people of Central America first learned to farm cacao plants around two thousand years ago. The Maya took cacao trees from the rainforests and grew them in their gardens. They cooked cacao seeds, then crushed them into a soft paste. They mixed the paste with water and flavourful spices to make an unsweetened chocolate drink. The Maya poured the chocolate drink back and forth between two containers so that the liquid would have a layer of bubbles or foam.

Cacao and chocolate were an important part of Maya culture. There are often images of cacao plants on Maya buildings and art objects. Ruling families drank chocolate at special ceremonies. And, even poorer members of society could enjoy the drink once in a while. Historians believe that cacao seeds were also used in marriage ceremonies as a sign of the union between a husband and a wife.

The Aztec culture in current-day Mexico also prized chocolate. But cacao plants could not grow in the area where the Aztecs lived. So, they traded to get cacao. They even used cacao seeds as a form of money to pay taxes. Chocolate also played a special role in both Maya and Aztec royal and religious events. Priests presented cacao seeds and offerings to the gods and served chocolate drinks during sacred ceremonies. Only the very wealthy in Aztec society could afford to drink chocolate because cacao was so valuable. The Aztec ruler Montezuma was believed to drink fifty cups of chocolate every day. Some experts believe the word for chocolate came from the Aztec word *xocolatl*, which in the Nahuatl language means “bitter water.” Others believe the word “chocolate” was created by combining Mayan and Nahuatl words.

C

The explorer Christopher Columbus brought cacao seeds to Spain after his trip to Central America in 1502. But it was the Spanish explorer Hernando Cortes who understood that chocolate could be a valuable investment. In 1519, Cortes arrived in current-day Mexico. He believed the chocolate drink would become popular with Spaniards. After the Spanish soldiers defeated the Aztec empire, they were able to seize the supplies of cacao and send them home. Spain later began planting cacao in its

colonies in the Americas in order to satisfy the large demand for chocolate. The wealthy people of Spain first enjoyed a sweetened version of the chocolate drink. Later, the popularity of chocolate spread throughout Europe. The English, Dutch and French began to plant cacao trees in their own colonies. Chocolate remained a drink that only wealthy people could afford until the eighteenth century. During the period known as the Industrial Revolution, new technologies helped make chocolate less costly to produce.

D

Farmers grow cacao trees in many countries in Africa, Central and South America. The trees grow in the shady areas of the rainforests near the Earth's equator. But these trees can be difficult to grow. They require an exact amount of water, warmth, soil and protection. After about five years, cacao trees start producing large fruits called pods, which grow near the trunk of the tree. The seeds inside the pods are harvested to make chocolate. There are several kinds of cacao trees. Most of the world's chocolate is made from the seeds of the *forastero* tree. But farmers can also grow *criollo* or *trinitario* cacao plants. Cacao trees grown on farms are much more easily threatened by diseases and insects than wild trees.

Growing cacao is very hard work for farmers. They sell their harvest on a futures market. This means that economic conditions beyond their control can affect the amount of money they will earn. Today, chocolate industry officials, activists and scientists are working with farmers. They are trying to make sure that cacao can be grown in a way that is fair to the farmers and safe for the environment.

E

To become chocolate, cacao seeds go through a long production process in a factory. Workers must sort, clean and cook the seeds. Then they break off the covering of the seeds so that only the inside fruit, or nibs, remain. Workers crush the nibs into a soft substance called chocolate liquor. This gets separated into cocoa solids and a fat called cocoa butter. Chocolate makers have their own special recipes in which they combine chocolate liquor with exact amounts of sugar, milk and cocoa fat. They finely crush this "crumb" mixture in order to make it smooth. The mixture then goes through two more processes before it is shaped into a mould.

Chocolate making is big business. The market value of the yearly cacao crop around the world is more than five billion dollars. Chocolate is especially popular in Europe and the United States. For example, in 2005, the United States bought 1.4 billion dollars' worth of cocoa products. Each year, Americans eat an average of more than five kilograms of chocolate per person. Speciality shops that sell costly chocolates are also very popular. Many offer chocolate lovers the chance to taste chocolates grown in different areas of the world.

Questions 1–5

Reading Passage 1 has five paragraphs, **A–E**.

Which paragraph contains the following information?

Write the correct letter, **A–E**, in boxes **1–5** on your answer sheet.

NB you can use any letter more than once.

- 1** the part of cacao trees used to produce chocolate
- 2** average chocolate consumption in the United States
- 3** risks faced by farmers in the cacao industry
- 4** where the first sweetened chocolate drink appeared
- 5** how ancient American civilizations obtained cacao

Questions 6–10

Do the following statements agree with the information given in Reading Passage 1?

In boxes **6–10** on your answer sheet, write:

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this @Teaxona

- 6** Use of cacao and chocolate in ceremonies was restricted to Maya royal families.
- 7** Hernando Cortes invested in chocolate and chocolate drinks.
- 8** The *forastero* tree produces the best chocolate.
- 9** Some parts of the cacao seed are removed during production.
- 10** Chocolate is more popular in some regions of the world than others.

Questions 11–13

The flow chart below shows the steps in chocolate making.

Complete the flow chart using **NO MORE THAN THREE WORDS** from the passage.

Write your answers in boxes **11–13** on your answer sheet.

Chocolate Production

Cacao seeds

- sorting, cleaning and cooking
- ridding seeds of their **11** _____

Nibs

- crushing
- add sugar, milk and **12** _____

Crumb mixture

- crush finely
- then shaped in a **13** _____

Test 20: The History of The Tortoise

If you go back far enough, everything lived in the sea. At various points in evolutionary history, enterprising individuals within many different animal groups moved out onto the land, sometimes even to the most parched deserts, taking their own private seawater with them in blood and cellular fluids. In addition to reptiles, birds, mammals and insects which we see all around us, other groups that have succeeded out of water include scorpions, snails, crustaceans such as woodlice and land crabs, millipedes and centipedes, spiders and various worms. And we must not forget the plants, without whose prior invasion of the land none of the other migrations could have happened.

Moving from water to land involved a major redesign of every aspect of life, including breathing and reproduction. Nevertheless, a number of land animals later reversed direction, abandoned their hard-earned terrestrial adaptations, and returned to the water again. Seals have only gone part way back. They show us what intermediates might have been like on the way to extreme cases such as whales and dugongs. Whales (including dolphins) and dugongs, with their close cousins the manatees, ceased to be land creatures altogether and reverted fully to marine habits. They do not come ashore to breed, but they still breathe air, having never developed anything resembling the gills of their earlier marine ancestors. Turtles returned to the sea a very long time ago and, like all vertebrate returnees to the water, they breathe air. However, in one respect, they are less completely returned to the water than whales or dugongs, for turtles still lay their eggs on beaches.

There is evidence that all modern turtles are descended from a terrestrial ancestor which lived before most dinosaurs. Two key fossils—*Proganochelys quenstedti* and *Palaeochersis talampayensis*—date from early dinosaur times and appear to be close to the ancestry of all modern turtles and tortoises. You might wonder how we can tell whether fossil animals lived on land or in water, especially if only fragments remain. Sometimes it is obvious. Ichthyosaurs were reptilian contemporaries of the dinosaurs, with fins and streamlined bodies. The fossils look like dolphins and they surely lived like dolphins, in the water. With turtles it is a little less obvious. One way to tell is by measuring the bones of their forelimbs.

Walter Joyce and Jacques Gauthier, at Yale University, obtained three measurements in these particular bones of 71 species of living turtles and tortoises. They used a kind of triangular graph paper to plot the three measurements against one another. All the land tortoise species formed a tight cluster of points in the upper part of the triangle; all the water turtles clustered in the lower part. There was no overlap, except with species that spend time both in water and on land. These amphibious species showed up on the triangular graph roughly halfway between the “wet cluster” of sea turtles and the “dry cluster” of land tortoises. The next step was to determine where the fossils fell. The

points for *P. quenstedti* and *P. talampayensis* lie firmly within the dry cluster. Both were dry-land tortoises. They lived before turtles returned to the water.

You might think, therefore, that modern land tortoises have remained terrestrial ever since those early times. But this is not so. If you draw out the family tree of all modern turtles and tortoises, nearly all the branches are aquatic. Today's land tortoises form just a single branch, deeply nested among aquatic branches. This suggests that modern land tortoises have not been exclusively terrestrial since the era of *P. quenstedti* and *P. talampayensis*. Rather, their ancestors went back to the water, and they then re-emerged onto land in more recent times.

Tortoises therefore represent a remarkable double return. Like mammals, reptiles and birds, their remote ancestors were marine fish and, before that, more primitive creatures stretching back to primeval bacteria. Later ancestors lived on land for countless generations. Later still, some evolved back into the water and became sea turtles. Finally, they returned once again to the land as tortoises, some now living in the driest of deserts.

***Question 14**

Choose the correct letter, **A, B, C or D**.

Write the correct letter in box **14** on your answer sheet.

According to the writer, the most significant thing about tortoises is that

- A** they can live in extremely dry environments.
- B** their earliest ancestor was a type of primitive bacteria.
- C** they share many similarities with sea turtles.
- D** they have moved from sea to land more than once.

Questions 1–4

Answer the questions below.

Choose **NO MORE THAN THREE WORDS** from the passage for each answer.

Write your answers in boxes **1–4** on your answer sheet.

- 1** What had to move onto land before any animals could migrate there?
- 2** What **TWO** aspects of life had to change significantly for animals to live on land?
- 3** What feature from their ancestors do whales no longer have?
- 4** What animals might ichthyosaurs have looked like?

Questions 5–7

Do the following statements agree with the information in Reading Passage 1?

In boxes **5–7** on your answer sheet, write:

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this @Tea

- 5** Turtles were among the first animals to return to the sea.
- 6** When fossil remains are incomplete, it is always difficult to determine the animal's habitat.
- 7** The habitat of ichthyosaurs can be identified from the appearance of their fossils.

Questions 8–13

Complete the flow chart below.

Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

Write your answers in boxes **8–13** on your answer sheet.

Method of determining where the ancestors of turtles and tortoises came from

Step 1

71 species of living turtles and tortoises were examined, and a total of
8 _____ were taken from the bones of their forelimbs.

Step 2

The data was recorded on a
9 _____,
used to compare the measurements.

Outcome:

Land tortoises formed a dense
10 _____
of points near the top, while sea turtles were grouped at the bottom.

Step 3

The same data was collected from some living
11 _____
species and added to the results.

Outcome:

These species appeared about
12 _____
up the triangular graph between the land tortoises and sea turtles.

Step 4

Bones of *P. quenstedti* and *P. talampayensis* were analysed similarly.

Outcome:

The position of the points showed that both ancient creatures were
13 _____.

Test 21: A History of Fingerprinting

A

To detectives, the answers lie at the end of our fingers. Fingerprinting offers an accurate and reliable means of personal identification. The ability to identify a person from a mere fingerprint is a powerful tool in the fight against crime and is the most commonly used type of forensic evidence. Although traditional ink methods have largely been replaced by faster digital techniques, the basic principles remain unchanged: every person's fingerprints are unique. But what makes our fingerprints different from those of everyone else?

B

A good place to start is to understand what fingerprints are and how they are formed. A fingerprint is the arrangement of ridges and furrows on the tips of the fingers. These patterns develop completely during foetal growth and remain unchanged throughout a person's life. Whereas characteristics such as height and weight change over time, fingerprints do not. Their uniqueness arises because genetic factors combine with small environmental influences in the womb, such as temperature, causing the ridges to grow at varying rates. Even identical twins do not share the same fingerprints.

C

It is difficult to identify the exact moment in history when the distinctive nature of fingerprints was first recognised. Evidence suggests that ancient civilisations, including those in Babylon and China, used thumbprints on clay tablets to authorise business transactions. Whether these marks were understood as personal identifiers or simply symbolic gestures is unknown.

D

Nevertheless, several individuals made major contributions to the study of fingerprints. A 14th-century Persian physician is believed to have been one of the earliest to state that no two fingerprints were alike. Later, in the 17th century, Marcello Malpighi examined the shapes of loops and spirals in fingerprints, and a layer of skin was later named after him. It was William Herschel, an officer with the East India Company, who grasped the full potential of fingerprints. He used them as signatures on contracts to prevent fraud and spent two decades studying them, concluding that fingerprints were both unique and permanent. In 1880, Henry Faulds proposed their use in criminal investigations and contacted Charles Darwin, whose cousin Francis Galton later published the first major scientific study on the topic.

E

Even though the uniqueness of fingerprints had long been known, they were not widely used for criminal identification until the early 20th century. Earlier societies relied on branding, tattooing or maiming criminals. Roman authorities tattooed soldiers to prevent desertion, while other cultures marked criminals visibly so they could be easily recognised.

F

For years, Western police forces resisted fingerprinting, preferring the Bertillon System, which relied on taking measurements of various body parts. This system collapsed in 1903 when two prisoners at Leavenworth Penitentiary—Will West and William West—were found to have virtually identical measurements. Only their fingerprints distinguished them, and from then onwards fingerprinting became the standard method of identification.

G

While fingerprinting proved useful for identifying criminals already on record, it was less helpful in cases where no suspect appeared in the database. Recently, fingerprint chemistry has expanded what can be learned from prints. Traces of organic compounds can reveal whether the person was a child, a smoker or even their approximate age. After centuries of use, fingerprints still offer new possibilities.

Questions 1–6

Reading Passage 1 has seven paragraphs, A–G.

Choose the correct heading for **paragraphs B–G** from the list of headings below.

Write the correct number, **i–x**, in boxes **1–6** on your answer sheet.

The list of heading

- | | |
|-------------|--|
| i | Key people who changed the study of fingerprints |
| ii | An alternative to fingerprinting |
| iii | The importance of fingerprints |
| iv | How criminals used to be identified |
| v | Development of fingerprint patterns |
| vi | Family connections |
| vii | Recent scientific progress |
| viii | A surprising coincidence |
| ix | Punishments for criminals |
| x | An unclear history |

Example

Paragraph A → **iii**

1. Paragraph B
2. Paragraph C
3. Paragraph D
4. Paragraph E
5. Paragraph F
6. Paragraph G

Questions 7–9

Complete the sentences below.

Choose **NO MORE THAN TWO WORDS** from the passage for each answer.

Write your answers in boxes 7–9 on your answer sheet. @Teaxona

7. Unlike other _____ that can change, fingerprints remain the same for life.
8. Although genetically identical, _____ do not share the same fingerprints.
9. In India, a fingerprint was used instead of a _____ on contracts.

Questions 10–14

Do the following statements agree with the information given in Reading Passage 1?

In boxes 10–14 on your answer sheet, write:

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this @Teaxona

10. Fingerprinting is the only effective method for identifying criminals.
11. Fingerprint patterns are formed before a baby is born.
12. Malpighi's work took place in Italy.
13. Roman soldiers were tattooed to deter them from violent behaviour.
14. Fingerprint chemistry can show whether a print belongs to an older person.

Test 22: The Can – A Brief History Lesson

A

The story of the can begins in 1795 when Nicholas Appert, a Parisian, had an idea: why not pack food in bottles like wine? Fifteen years later, after researching and testing his idea, he published his theory: if food is sufficiently heated and sealed in an airtight container, it will not spoil. In 1810 Peter Durand, an Englishman, wanted to surpass Appert's invention, so he elected to try tin instead of glass. Like glass, tin could be sealed airtight but tin was not breakable and was much easier to handle. Durand himself did no canning, but two other Englishmen, Bryan Donkin and John Hall, used Durand's patent. After experimenting for more than a year, they set up a commercial canning factory and by 1813 they were sending tins of food to British army and navy authorities for trial.

B

Perhaps the greatest encouragement to the newborn canning industry was the explosion in the number of new colonial territories. As people and goods were being transported to all parts of the world, the can industry itself was growing in new territories. Englishmen who emigrated to America brought their newfound knowledge with them. One of these was Thomas Kensett, who might fairly be called the father of the can manufacturing industry in the United States. In 1812 he set up a small plant on the New York waterfront to can the first hermetically sealed products in the United States.

C

Just before the Civil War, a technical advance by canners enabled them to speed up production. Adding calcium chloride to the water in which cans were cooked raised the water temperature, speeding up the canning process. Also for almost 100 years, tin cans were made by artisans by hand. It was a laborious process, requiring considerable skill and muscle. As the industrial revolution took hold in the United States, the demand for cans increased and machines began to replace the artisans' handiwork. A good artisan could make only 10 cans a day. True production progress in can-making began in 1922, when American engineers perfected the body-making process. New methods soon increased production of cans to as many as 250 a minute.

D

As early as 1940, can manufacturers began to explore the possibility of adapting cans to package carbonated soft drinks. The can had to be strengthened to accommodate higher internal can pressures created by carbonation (especially during warm summer months), which meant increasing the thickness of the metal used in the can ends. Another concern for the new beverage can was its shelf life. Even small amounts of dissolved tin or iron from the can could impair the drinking quality of drinks. Also the food acids, including carbonic, citric and phosphoric, in soft drinks presented a risk for the rapid corrosion of exposed tin and iron in the can. At this point the can was upgraded by improving the organic coatings used to line the inside. The can manufacturers then embarked on a program of material and cost savings by

reducing both the amount of steel and the amount of coating used in can making. These efforts were in part inspired by a new competitor — aluminum.

E

Beverage cans made from aluminum were first introduced in 1965. This was an exciting innovation for the packaging industry because the aluminum can was made with only two pieces — a body and an end. This made production easier. Some of the reasons for the aluminum can's acceptance were its ductility, its support of carbonation pressure, its lighter weight and the fact that aluminum does not rust. Both steel and aluminum cans used an easy-open end tab but the aluminum tab was much easier to make. Perhaps the most critical element in the aluminum can's market success was its recycling value. Aluminum can recycling excelled economically in the competition with steel because of the efficiencies aluminum cans realized in making new cans from recycled materials compared with 100 percent virgin aluminum. Steel did not realize similar economies in the recycling process.

F

Prior to 1970, can makers, customers and consumers alike were unaware of the impact that the mining and manufacturing of steel or aluminum had on the environment. The concept of natural resource preservation was not an issue of great importance and the low growth of population during these early years further de-emphasized concerns for resource depletion. Both industries, however, came to realize the importance of reducing their impact on the environment in the late 1960s and early 1970s as a new environmentally conscious generation emerged. Manufacturers began to recognize the economics of recycling, namely lower manufacturing costs from using less material and less energy. By the 1980s and 1990s, recycling had become a way of life. Aluminum can recycling has become a billion-dollar business and one of the world's most successful environmental enterprises. Over the years, the aluminum can has come to be known as America's most recyclable package, with over 60 percent of cans being recycled annually.

G

Advances in can manufacturing technology have also brought us lighter aluminum cans. In 1972, one pound of aluminum yielded only 21.75 cans. Today, by using less material to make each can, one pound of aluminum makes approximately 32 cans — a 47 percent improvement. Just the lightening of can ends makes a huge difference. When you multiply the savings by the 100 billion cans that are made each year, the weight and savings are phenomenal — over 200 million pounds of aluminum!

Questions 1–6

Choose the correct heading for paragraphs B–G.

Write the correct **i–xi** in boxes **1–6**.

List of Headings

- i The Innovation
- ii Technological Breakthroughs
- iii Canning and the Beer Industry
- iv The Invention
- v Canning and War
- vi Further Manufacturing Progress
- vii Problems with Spoiled Contents
- viii Expansion of the Industry
- ix Today's Uses for Canning
- x Drinks Canning
- xi Cans and the Environment

Example

Paragraph A → **iv**

1. Paragraph B
2. Paragraph C
3. Paragraph D
4. Paragraph E
5. Paragraph F
6. Paragraph G

Questions 7–12

Match the date with the correct event.

Write the correct letter **A–G** in boxes **7–12**.

List of events

- A Mass production techniques revolutionized the canning process.
- B Tinned food was tested by military authorities.
- C Today's canning material was first introduced.
- D The first American canning factory was opened.
- E Tin was used in the canning process for the first time.
- F The canning of fizzy drinks began.
- G The first business canning plant was opened.

List of dates

- 7. 1922
- 8. 1812
- 9. 1813
- 10. 1965
- 11. 1940

Questions 12–14

*Do the following statements agree with the information given in Reading Passage 1?
In boxes **12–14** on your answer sheet, write:*

TRUE *if the statement agrees with the information*

FALSE *if the statement contradicts the information*

NOT GIVEN *if there is no information on this @Teaxona*

- 12. Recycling has helped reduce manufacturing overheads.
- 13. The weight of the aluminum cans had increased compared to the past.
- 14. Aluminium can production costs have fallen by nearly 50% since 1972.

Test 23: The History of The Invention of Plastics

A

Natural polymers include such familiar substances as silk, rubber and cotton. Plastics are artificial polymers. Plastics are used on a daily basis throughout the world. The word plastic is a common term used for many materials of a synthetic or semi-synthetic nature. The term was derived from the Greek *plastikos*, which means “fit for moulding.” Plastics are a wide variety of combinations of properties when viewed as a whole. They are used for shellac, cellulose, rubber and asphalt. We also synthetically manufacture items such as clothing, packaging, automobiles, electronics, aircraft, medical supplies and recreational items. Much of what we have today would not be possible without plastics.

B

In the early part of the twentieth century, a big boom occurred in polymer chemistry when polymer materials such as nylon and Kevlar appeared. Much of the work done with polymers focuses on improvement while using existing technologies, but chemists also have opportunities ahead. There is a need for new applications for polymers, always looking for less expensive materials that can replace what is used now. Chemists must also be more aware of what the market demands, such as environmentally friendly polymers that break down safely. Concerns such as these have brought new energy to the science arena, and there are always new discoveries to be made.

C

The evolution of plastic chemistry is complex, and the uses for plastics are endless. In the Middle Ages, when scientists first began to experiment, plastics were derived from natural organic sources such as egg and blood proteins. It was not until the 19th and 20th centuries that the plastics we know today were created. Charles Goodyear began the modern plastic revolution when he vulcanised rubber in 1839, paving the way for the tyre. Prior to this discovery, rubber products did not hold up well in warm temperatures. Rubber was only one source of plastic, however, and three key inventors followed Goodyear’s path by turning plastic from a hard, nearly unusable substance into the invaluable material it is today.

D

Alexander Parkes, born in Birmingham, England in 1813, was the son of a brass lock manufacturer. Raised around metal fabrication, he worked first as an apprentice in Birmingham’s brass foundry owned by Samuel S. Messengers and Sons. Parkes later shifted his focus to electroplating when he worked for George and Henry Elkington. It was there that Parkes developed his inventive spirit. His first patent, awarded in 1841,

dealt with electroplating delicate items such as flowers. Throughout his career, Parkes held more than 80 patents in metals and plastics. He is credited with inventing the first man-made plastic, Parkesine, in 1856. Parkes introduced this combination of nitrocellulose and solvents to England in 1862 at the London International Exhibition. Parkesine was introduced in 1862, but its original formulation was too flammable to be successful. However, it laid the foundation for future inventors, including John Wesley Hyatt.

E

John Wesley Hyatt, born in Starkey, New York in 1837, patented several hundred inventions. His link to plastics began with the game of billiards. Billiard balls were originally made of ivory, which became scarce in the 1800s. Attracted by a \$10,000 reward, Hyatt attempted to create a substitute material. Using Parkes' solid nitrocellulose and Frederick Scott Archer's discovery of liquid nitrocellulose, Hyatt developed celluloid, patented in 1870.

F

Celluloid was used for numerous products, including billiard balls, false teeth, combs, baby rattles and piano keys. Despite its replacement in many modern industries, Hyatt's version of celluloid is still used to produce ping-pong balls. The invention of celluloid was a crucial step in the manufacturing of plastics and played a role in early film production.

G

Hyatt's work later influenced Leo Baekeland, a Belgian-born chemist. Baekeland's first major invention was Velox, a photographic paper that allowed photographs to be taken in artificial light. George Eastman, founder of Eastman Kodak, purchased the Velox process from Baekeland for \$750,000 in 1899. Baekeland used that money to fund his own home laboratory.

H

Baekeland progressed from photographic paper to synthetic resins and invented Bakelite in 1907, made from phenol and formaldehyde. Bakelite was patented in 1909 and became known as the product that led the world into the Age of Plastics. It was used in items ranging from buttons and furniture to early television sets. Bakelite is still used today in products such as car brakes and space shuttle components.

Questions 1–5

Use the information in the passage to match the inventors, chemists or companies (listed **A–F**) with the statements below.

Write the correct letter **A–F** in boxes **1–5** on your answer sheet.

- A** Alexander Parkes
- B** Charles Goodyear
- C** John Wesley Hyatt
- D** Frederick Scott Archer
- E** Leo Baekeland
- F** George Eastman

- 1** invested a significant amount of money in a process that greatly benefited his company
- 2** produced a formula successful only experimentally, but significant for future progress
- 3** maintained a laboratory using his own financial resources
- 4** created a substitute material after being motivated by a financial reward
- 5** founded a business associated with automotive parts

Questions 6–9

Do the following statements agree with the information in Reading Passage 1?

In boxes **6–9** on your answer sheet, write:

- TRUE** if the statement agrees with the information
- FALSE** if the statement contradicts the information
- NOT GIVEN** if there is no information on this @Teaxona

- 6** Silk, cotton and plastics share a similar characteristic.
- 7** Chemists' goals sometimes conflict with market expectations.
- 8** Most of Parkes' patents were in only two main fields.
- 9** Celluloid is no longer used at all due to new synthetic materials.

Questions 10–14

Complete the summary below.

*Choose **NO MORE THAN THREE WORDS** from the passage for each answer.*

*Write your answers in boxes **10–14** on your answer sheet.*

Alexander Parkes, a well-known chemist, grew up surrounded by **10** _____ in an industrial city.

He began his career in a **11** _____ in his hometown and later developed his **12** _____ while working for other employers.

He went on to earn more than 80 patents in the field of **13** _____.

Although his most important invention, Parkesine, made from **14** _____, was considered the first artificial plastic, it did not achieve much success. However, it paved the way for later scientists.

Test 24: The History of Ice Cream

People the world over love ice cream. But they seldom consider how the frozen dessert came into being. Ice cold desserts, called 'ices', were around as far back as the 5th century BC, when the ancient Greeks refreshed themselves with snow, flavoured with honey or fruits. Although the Greek physician Hippocrates warned his patients to avoid unhealthy chilled drinks, these icy treats nonetheless enjoyed tremendous popularity. The ancient Romans were also familiar with ices. While the philosopher Seneca criticised the emperor Nero in the 1st century AD for the extravagance of having ice and snow brought from the mountains to cool his swimming pool, he seemed to have seen nothing wrong with the emperor's fondness for chilled drinks and snow flavoured with honey.

But at that time, unless Romans wanted to eat their ices only in winter or high in the mountains, they needed not only to bring snow or ice to where they consumed it, but also to find some way of storing it in warm weather. This required a special sort of building now known as an ice house. Usually constructed of stone, underground and insulated with straw, these were sufficiently cold to preserve ice cut from frozen lakes and rivers in winter, or snow carried down from the mountains. Ice houses were in regular use in ancient Rome, and their growing use soon led to the development of a busy trade in snow and ice crisscrossing Europe, the Mediterranean and parts of western Asia.

Yet desserts made with cold dairy products were not invented until much later. During the Tang Dynasty (AD 616–907), the Chinese made a cold, creamy concoction of buffalo milk, flour, camphor and snow. But while it was easy enough to cool milk down, this mixture proved difficult to freeze. This was simply a matter of physics. Most milk freezes at between -0.535 degrees C and -0.565 degrees C, which is below the freezing point of water; so packing it in frozen water or snow doesn't get milk cold enough to become solid. Surprisingly, the solution to this problem was salt. When salt is added to snow or ice, energy is absorbed from the container of milk, for example, set within the snow or ice, reducing the milk's temperature and creating a basic form of refrigeration or freezing.

No one knows precisely when a mixture of salt and snow or ice was first used to produce ice cream. The earliest recorded references to what we know as ice cream occurred in Europe around 1530. However, it was not until the publication of a cookbook by Mrs Mary Eales in 1718 that recipes showing how to make it appeared in English. At that time, there was considerable variation in the ingredients. Milk, cream and custard were used almost interchangeably; and cooks ignored the food recommended including, for example, rose petals, caramel, ginger, lemon, chocolate, parsley and even asparagus.

For many years, historians assumed that it was the aristocracy that were the main consumers of these early ice creams. During the 18th century, for example, a range of beautiful ice cream dishes were created for French King Louis XV's court. In 1813, US President James Madison's wife arranged for ice creams to be given to guests at her husband's inaugural ball; yet two years later, the French writer Balzac described the lavish ice cream desserts consumed in the elegant houses of Paris in his novel *Splendeurs et misères des courtisanes*. But, especially in 18th-century Naples in Italy, in part because of the relatively easy access to snow, it was also possible for people much lower down the social scale to make, sell and enjoy ice cream. Not only were there shops and cafés selling ice creams, but there were also vendors who sold it along the public roads, and there is evidence to suggest that some families may have made it at home.

However, early ice creams were still quite different from the iced treats we enjoy today. Since the milk, cream or custard was not stirred as it was frozen, it formed icy chunks, resulting in a lumpy, disagreeable texture. In 1843, however, an American called Nancy Johnson designed a hand-turned freezer, which allowed her to make silky smooth ice cream. Her invention consisted of a large container filled with ice and salt, a tightly closed cylinder for the milk or cream and a removable handle, which was rotated to stir the ice cream until properly frozen.

Even with Nancy Johnson's freezer, however, it was only possible to make small batches of ice cream at a time. But in 1926 another American, Clarence Vogt, improved on Johnson's idea with his continuous freezer, which made it possible to produce ice cream on an industrial scale. The inventions of Johnson and Vogt resulted in an explosion in the output of ice cream, which pushed the price down, making ice cream available to everyone everywhere. As more companies started manufacturing their own brands, increased competition led to the standardisation of the commercial product and to more inventive varieties.

NOTE: THIS IS A PRE-TEST (hence 16 questions)

Questions 1–8

Do the following statements agree with the information given in Reading Passage 1?

In boxes **1–8** on your answer sheet, write

TRUE if the statement agrees with the information

FALSE if the statement contradicts the information

NOT GIVEN if there is no information on this @Teaxona

1. The majority of ancient Greeks followed the advice of Hippocrates about cold foods.
2. Seneca had different reactions to Nero's various uses of snow.
3. Many ancient Romans went to the mountains in the summer to eat ice cream.
4. The cold storage houses in Rome were generally built of frozen blocks of ice.
5. The buying and selling of snow and ice became a common international activity during Roman times.
6. Tang Dynasty Chinese preferred chilled desserts made from milk to those made only of snow and ice.
7. The freezing temperature of most types of milk made it easy to create frozen dairy desserts.
8. Salt was commonly used as a flavouring in ice creams in the past.

Questions 9–16

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes **9–16** on your answer sheet.

The development of modern ice cream

Early publications about making ice cream

- 1530 – first known written references to ice cream
- 18th century
 - first book in English with **9** _____ for ice cream was published in 1718
 - many different **10** _____ were added to ice cream desserts

Notable consumers of ice cream

- it was once widely thought that ice cream was mostly eaten by the **11** _____
- ice cream desserts were
 - created for French King Louis XV's court
 - served at an important **12** _____ for an American president

18th-century Naples, Italy

- lower-class people also enjoyed ice cream
- **13** _____ of ice cream could be seen on the streets

Improvements in ice cream production

- before 1843, ice cream was not stirred while being frozen so it had an unpleasant **14** _____
- Nancy Johnson invented a freezer which included a **15** _____ to turn a cylinder @Teaxona
- 20th-century production methods allowed a reduction in the **16** _____ of commercially sold ice cream

T1	Pencil
1	Tree
2	Sheep
3	String
4	Government
5	Steal
6	Grease
7	True @Teaxona
8	Not Given
9	False
10	False
11	Not Given/False?
12	True
13	False

T4	Poster
1	Storage space
2	Pantograph
3	Color/colour
4	Greasy crayon
5	Transfer paper
6	Words @Teaxona
7	Communication
8	Exhibition
9	Cultural institutions
10	Not Given
11	True
12	True
13	False
14	

T2	Tea
1	Tombs
2	Book
3	Monks
4	Portuguese
5	Smuggling
6	India @Teaxona
7	clippers
8	False
9	Not Given
10	False
11	True
12	True
13	Not Given

T5	Glass
1	Obsidian
2	Spears
3	Beads
4	Impurities
5	Romans
6	Lead
7	Clouding
8	Taxes
9	True @Teaxona
10	False
11	Not Given
12	True
13	False

T3	Humans and Food
1	False
2	True
3	True
4	Not Given
5	False
6	Populations
7	Records
8	Salt @Teaxona
9	Glass
10	Texture
11	Light
12	Chisel
13	Meat

T6	Picnic
1	False
2	True
3	Not Given
4	True
5	Not Given
6	Not Given
7	True @Teaxona
8	Outside
9	Music
10	Cities
11	Aristocracy
12	Transport
13	Baskets

T7	Bicycles
1	100
2	Pedals
3	Toys
4	Overseas
5	Steel tubes
6	Greater
7	Better/safer bicycles
8	Chain
9	Rubber
10	False
11	True @Teaxona
12	Not Given
13	False

T10	Cakes
1	True
2	False
3	Not Given
4	True
5	False
6	True
7	Pastry
8	Ovens
9	Hobby
10	Wealth
11	Heavy
12	Columns
13	Design

T8	Guitar
1	False
2	True
3	Not Given
4	Not Given
5	True @Teaxona
6	False
7	Chartar
8	Body
9	Volume
10	America
11	Steel
12	Pickups
13	Gibson

T11	Cod Liver Oil
1	False
2	True
3	True @Teaxona
4	Not Given
5	Not Given
6	False
7	False
8	True
9	Wine
10	Coffee
11	Milk
12	Rope
13	Tail

T9	Chicken
1	True
2	Not Given
3	False
4	True
5	True
6	Not Given
7	False
8	Zoos
9	Tombs
10	Incubation
11	Humidity
12	Ovens
13	Predators

T12	Globe Theatre
1	False
2	True
3	Not Given
4	True
5	True
6	Not Given
7	Pit @Teaxona
8	Thatch
9	Tiring house
10	Musicians
11	Sprinklers
12	Half
13	Covered

T13	American Dance
1	Wood
2	Clay
3	Goddesses
4	Limbs
5	Wax @Teaxona
6	Porcelain
7	False
8	True
9	True
10	Not Given
11	True
12	False
13	Not Given

T16	Dolls
1	True
2	False
3	Not Given
4	False
5	Not Given
6	True @Teaxona
7	Walking
8	Book
9	Humour/Humor
10	Documentary
11	Lecturing
12	Comics
13	Mirrors

T14	Building Telegraph
1	True
2	False
3	Not Given
4	Not Given
5	True @Teaxona
6	Not Given
7	Expensive
8	Latex
9	Lead Pipe
10	Seaweed
11	President Buchanan
12	Camels
13	Route

T17	Salt
1	B
2	E
3	F
4	Essential element
5	Commercial application
6	Portable commodity
7	Tax @Teaxona
8	Spirits
9	True
10	Not Given
11	False
12	False
13	True

T15	Potato
1	False
2	False
3	Not Given
4	True
5	True
6	Flower
7	Prejudice
8	Reverse
9	Meat
10	Crops
11	Soil
12	Cultivation
13	Investment

T18	Rubber
1	G
2	F
3	B
4	E
5	D
6	C
7	F @Teaxona
8	Gas mileage
9	Radial tire
10	Natural rubber
11	Embargo
12	Blight
13	Vulnerable

T19	Cacao
1	D
2	E
3	D
4	C
5	B
6	False
7	Not given
8	Not given
9	True @Teaxona
10	True
11	Covering
12	Cocoa fat
13	Mould

T22	The Can
1	VIII
2	VI
3	X
4	I
5	XI
6	II
7	A
8	D
9	B
10	C
11	F
12	True @Teaxona
13	Not Given
14	False

T20	Tortoise
1	Plants
2	Breathing and reproduction
3	Gills
4	Dolphins
5	Not Given
6	False
7	True @Teaxona
8	Three measurements
9	(triangular) graph &/ paper
10	Cluster
11	Amphibious
12	Halfway
13	Dry-land tortoises
14	D

T23	Invention of Plastics
1	F
2	A
3	E
4	C
5	B
6	True
7	Not Given
8	True @Teaxona
9	False
10	Metal Fabrication
11	Brass Foundry
12	His Invention Spirit
13	Metals And Plastics
14	Nitrocellulose And Solvents

T21	Fingerprinting
1	V
2	X
3	I
4	IV
5	II / VIII
6	VII
7	(human) characteristics
8	Twins
9	Signatures
10	Not Given
11	True @Teaxona
12	Not Given
13	False
14	True

T24	Ice Cream
1	False
2	True
3	Not Given
4	False
5	True
6	Not Given
7	False
8	Not Given
9	Recipes
10	Ingredients
11	Aristocracy
12	Ball
13	Vendors
14	Texture
15	Handle
16	Price

